

# Multi-objective weighted average algorithm: a novel algorithm for multi-objective optimization problems and its application in engineering problems

Jun Cheng , Wim De Waele <sup>\*</sup> 

Laboratory Soete, Faculty of Engineering and Architecture, Department of Electromechanical, Systems and Metal Engineering, Ghent University, 9052, Zwijnaarde, Belgium

## ARTICLE INFO

### Keywords:

Multi-objective weighted average algorithm (MOWAA)  
Weighted average algorithm (WAA)  
Engineering problem  
Artificial neural network (ANN)  
Wire arc additive manufacturing (WAAM)

## ABSTRACT

Numerous meta-heuristic algorithms struggle with degraded performance when addressing multi-objective optimization problems due to the challenge of balancing two goals: accurately estimating Pareto-optimal solutions and ensuring their broad distribution across objectives. While the Weighted Average Algorithm (WAA) excels in single-objective optimization, its scalarization-based mechanism fundamentally conflicts with multi-objective requirements. To bridge this gap, we propose the Multi-Objective Weighted Average Algorithm (MOWAA) with three key innovations: (1) a hybrid exploration-exploitation mechanism integrating adaptive mutation and crossover operations; (2) an elitist archive management system using efficient non-dominated sorting across three critical solution sets; and (3) a novel roulette-wheel-based leader selection strategy that dynamically balances convergence and diversity. To verify the performance of the developed MOWAA, the numerical benchmark test functions (CEC2009, ZDT and DTLZ) and four engineering problems (the Binh and Korn (BNH), Constraint (CONSTR), Srinivas and Deb (SRN), and 10-bar Truss (BAR TRUSS)) are used in comparison with three multi-objective optimization algorithms. The results show that MOWAA achieves better optimization performance than comparative algorithms, with Pareto-optimal solutions exhibiting excellent convergence and coverage. Finally, applying MOWAA to an Artificial Neural Network (ANN) model using an experimental dataset on surface waviness (in mm) of Wire Arc Additive Manufacturing (WAAM) components enhances predictive accuracy by balancing optimization of prediction error and variance. Compared to single-objective optimization methods, the MOWAA approach effectively captures the complex relationships between process parameters and waviness in the WAAM process.

## 1. Introduction

The ever-increasing number of problems with more than one objective in real-world mechanical, civil and other engineering problems has created a significant demand for effective optimization algorithms (Zapotecas-Martínez et al., 2023). Multi-objective optimization is a popular field that focuses on addressing multiple, often conflicting, objectives to support decision making. Unlike a single objective optimization problem, multi-objective optimization seeks to find a set of solutions known as non-dominated or Pareto optimal solutions (Pereira et al., 2022; Zhang et al., 2022). A solution is deemed Pareto-optimal when no other feasible solution can improve one objective without compromising at least one of the others. These Pareto-optimal solutions

represent the optimal trade-offs among competing objectives, providing decision-makers with a spectrum of choices that reflect the inherent complexity of the problem. Notably, these solutions are identified without necessarily considering decision-makers' preferences during the optimization process, ensuring that the trade-offs are presented in a neutral and comprehensive manner.

Multi-objective optimization algorithms are generally categorized into two main approaches: a priori and a posteriori methods (Gunantara, 2018; Sharma and Kumar, 2022). Priori methods incorporate decision-makers' preferences at the outset, tailoring the optimization process to prioritize specific objectives or trade-offs. While these methods can streamline the solution process, they rely heavily on the accuracy and clarity of the preferences provided. Conversely, posteriori methods

<sup>\*</sup> Corresponding author.

E-mail address: [wim.dewaele@ugent.be](mailto:wim.dewaele@ugent.be) (W. De Waele).

<https://doi.org/10.1016/j.engappai.2025.111569>

Received 16 January 2025; Received in revised form 8 May 2025; Accepted 18 June 2025

Available online 2 July 2025

0952-1976/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

generate a diverse set of Pareto-optimal solutions, allowing decision-makers to evaluate the trade-offs and select their preferred solution post-optimization. This approach offers greater flexibility and insight into the problem's solution landscape, although it often comes at the cost of increased computational effort.

With regard to the former category, the multi-objective optimization problem is transformed into a single objective problem through assigning a weight to each objective and using a non-linear optimization method to combine each objective into a single objective (Li et al., 2022; Ma et al., 2020). For instance, Ma et al. (2020) reviewed the weight vector adjustment methods for decomposition-based multi-objective evolutionary algorithms. Among these, the majority of decomposition methods are based on the weighted sum (Mardle and Miettinen, 2000), the Tchebycheff method (Hu et al., 2024), and the penalty-based boundary intersection (Qi et al., 2020). Besides, based on the source of the heuristic information used in the adjustment, the weight vector adjusting strategies can be classified into random/predefined (Qin et al., 2023), fitting-based (Li et al., 2025), local-population-guided (De Farias et al., 2018), local-archive-guided (Wang et al., 2023), neighbor-weight-vector-guided (Li and Landa-Silva, 2008), and preference-based adjustments (Chen et al., 2021). However, due to the challenges in determining the relative importance of each objective during the optimization process, algorithms from the a priori category struggle to provide an effective weight adjustment strategy during the iteration process, often leading to unsuccessful outcomes in multi-objective optimization problems.

In the a posteriori category, the algorithms generate a set that contains all non-dominated solutions, or several of the well distributed non-dominated solutions, allowing decision makers to select the solution that best fits their needs. Compared to the a priori category, there is no need to determine the weight of each objective function during the optimization process (Deb, 2012). This approach is particularly well-suited to complex, real-world scenarios where preferences are uncertain or subject to change, offering both flexibility and a deeper understanding of the trade-offs inherent to the problem.

Based on their source of inspiration, the main multi-objective optimization algorithms can be classified into three different categories: 1) mathematics inspired; 2) evolution inspired; and 3) swarm intelligence inspired. As for the evolution inspired algorithms, the first algorithm used for solving a multi-objective optimization problem was named the vector evaluated genetic algorithm (VEGA) (Schaffer, 1985). VEGA utilizes  $m$  subpopulations to optimize each objective independently through proportional selection. Subsequently, these subpopulations are combined to undergo crossover and mutation operations. Following, Yen et al. (Yen and Lu, 2003) proposed an evolutionary approach based on the dynamic population size, named dynamic multi-objective evolutionary algorithm (DMOEA). However, its great demand for computation time limited its further application especially when the population size changed dynamically. To overcome this problem, Tan et al. (2006) proposed a distributed cooperative evolutionary algorithm that shared the computational workload among a group of computers over a network to reduce the computation time. Besides, to avoid the high computational complexity and non-elitism approach, the non-dominated sorting genetic algorithm II (NSGA-II) proposed in (González et al., 2022) used non-dominated sorting and congestion distance mechanisms, as well as suitable selection, crossover and mutation operations to effectively search Pareto optimal frontier for multi-objective optimization problems. To enhance the performance in solving up to 15 objectives, a reference point was introduced to select the optimal front in the extended algorithm called NSGA-III (Deb and Jain, 2014).

Concerning swarm based algorithms, the most popular algorithm is the multi-objective particle swarm optimization algorithm (MPSO) (Armano and Farmani, 2016), which used an alternative repository of particles to help other particles in their own flight. To improve the optimization performance, a local search method was integrated with

particle swarm optimization to produce more solutions around the personal best in the niching particle swarm optimization algorithms. Besides, Fei et al. (Ming et al., 2024) proposes a novel CMOEA, CMORWMDP, incorporating heterogeneous operators and normalization-based fitness, which demonstrates superior performance in solving multi-objective optimization problems. Other similar algorithms include rank-based multimodal immune algorithm (RMIA) (González et al., 2022), multi-objective grasshopper optimization algorithm (MOGOA) (X. Wang et al., 2023), multi-objective grey wolf optimizer (MOGWO) (Mirjalili et al., 2016), multi-strategy multi-objective artificial bee colony (MMABC) algorithm (D. M. Qin et al., 2023) and so on.

The last category comprises mathematics inspired algorithms. The most popular algorithm is the multi-objective chaos game optimization (MOCGO) (Khodadadi et al., 2023), which makes use of a leader selection methodology to strengthen its capabilities and to avoid the drawbacks of the original CGO method (Talatahari and Azizi, 2020), as well as an archive method to save non-dominated solutions. Recently, the decomposition concept and the random learning mechanism were incorporated into the MOCGO to generate several new versions (Yacoubi et al., 2023) including MOCGO/A, MOCGO/CD, MOCGO/D, and MOCGO/DR. In the MOCGO/D, the decomposition step employs a Normalized Boundary Intersection (NBI) technique for decomposing the multi-objective problem into single-objective sub-problems.

However, numerous meta-heuristic and evolutionary algorithms face significant challenges in maintaining robust performance when addressing multi-objective optimization problems. This is primarily due to the difficulty of balancing two conflicting goals: accurately estimating the Pareto-optimal solutions while ensuring their broad distribution across all objectives. While existing approaches, such as NSGA-II, MOEA/D, and SPEA2, have made notable contributions, many still rely on incremental modifications of classical frameworks, leading to diminishing returns in performance gains.

This limitation aligns with the well-known No Free Lunch (NFL) theorem (Wolpert and Macready, 1997), which states that no single optimization algorithm can perform efficiently across all possible problems. This principle holds true for both single-objective and multi-objective optimization approaches, underscoring the necessity of continuous improvement. To enhance optimization performance, two primary research directions have emerged: modifying existing well-established algorithms or developing novel optimization techniques. In previous research, the recently developed Weighted Average Algorithm (WAA) (Cheng and Waele, 2024a) has demonstrated considerable promise in solving single-objective optimization problems, thanks to its effective search capabilities. However, its strong performance in single-objective optimization stems from a fundamentally different paradigm; one that inherently prioritizes convergence to a single global optimum rather than balancing multiple competing objectives. This structural mismatch reveals a deeper theoretical gap: single-objective algorithms like WAA lack the necessary framework to define, let alone optimize, trade-offs between conflicting objectives. Unlike dedicated multi-objective approaches (e.g., Pareto dominance-based methods), WAA's weighted averaging mechanism collapses multiple objectives into a scalar value, inherently suppressing the diversity required for Pareto-optimal solutions.

This limitation is not unique to WAA but exemplifies a broader challenge: single-objective algorithms cannot be trivially extended to MOOPs without sacrificing their core principles. While scalarization techniques (e.g., weighted sum) may superficially bridge this gap, they often fail to capture the true Pareto front, especially for non-convex or discontinuous landscapes. Therefore, merely "enhancing" WAA for MOOPs is insufficient; a paradigm shift from single-point convergence to population-driven, diversity-preserving search mechanisms is needed.

In light of these considerations, researchers proposed several transformation frameworks that extend single-objective algorithms, such as Heat Transfer Search and the Weighted Aggregation Approach, into

robust multi-objective optimizers. This is achieved by integrating problem-specific mechanisms such as decomposition strategies (Kumar et al., 2022), crossover-based reproduction phases (Kumar et al., 2021a), and hybridized search modules (Kumar et al., 2021b). These enhancements collectively aim to overcome the inherent limitations of traditional single-objective models by enabling more accurate convergence toward Pareto-optimal fronts, improved solution diversity, and a balanced distribution across the objective space. However, existing strategies still suffer from limited generalization capability, high sensitivity to parameter settings, and a lack of theoretical convergence guarantees. Their performance may also degrade when tackling high-dimensional objectives, dynamic constraints, or complex real-world problems.

To address these challenges, the current research proposes an improved transformation framework that adapts single-objective algorithms into the multi-objective domain. In particular, the algorithm WAA is extended by incorporating mutation and crossover operators into its exploitation process, significantly enhancing its ability to explore the objective space, to estimate Pareto-optimal solutions more accurately, and to maintain diversity across the solution set.

This represents a concerted effort to strike an effective balance between exploration and exploitation, thus advancing the state-of-the-art in multi-objective optimization. This so-called multi-objective weighted average algorithm (MOWAA) makes use of different strategies (including a leader selection, a specific archive solution set updating strategy, and a fixed size external archive solution set) to strengthen WAA's capacities for solving multi-objective optimization problems. To evaluate the optimization performance, the proposed MOWAA is applied to a wide range of problems including unconstrained benchmark test functions (CEC2009, ZDT and DTLZ) and constrained engineering problems. Besides, several state-of-the-art multi-objective optimization algorithms are compared with the proposed MOWAA on the basis of several performance metrics including Inverted Generational Distance (IGD), Generational Distance (GD), Spacing (S), Maximum Spread (MS) and hypervolume (HV). The results will demonstrate that the proposed MOWAA has a superior ability to handle multi-objective optimization problems. Besides, through the comparison of the ideal Pareto-front solution and the obtained Pareto-front solution, excellent convergence and convergence rate will be demonstrated. Moreover, machine learning continues to see widespread application across various domains. For example, recent studies (Mohamed et al., 2024; Srikanth et al., 2024; Yue et al., 2024) highlight its role in marine ecosystem analysis, FRP-concrete behavior prediction, and structural load capacity modeling. However, these applications also reveal ongoing challenges, including limited model interpretability, generalization ability, and sensitivity to input features—underscoring the importance of integrating optimization algorithms to improve robustness and predictive performance. To this end, MOWAA was further integrated with an Artificial Neural Network (ANN) to construct a robust surrogate model for predicting surface waviness of Wire Arc Additive Manufactured components. Unlike the WAA-ANN model, which primarily focuses on minimizing the root mean square error (RMSE), the MOWAA-ANN model simultaneously optimizes both RMSE and the coefficient of determination (R), achieving a balanced improvement in accuracy and model fit. Comparative analysis demonstrates that MOWAA-ANN significantly outperforms both WAA-ANN and standalone ANN models, as evidenced by lower RMSE values and higher R-squared scores. This highlights the effectiveness of multi-objective optimization in enhancing the predictive power and generalization capability of

machine learning models in complex engineering applications.

The original contributions of this paper are as follows.

- 1) Proposal of a Multi-Objective Weighted Average Algorithm (MOWAA): The single-objective Weighted Average Algorithm (WAA) is extended and transformed into a multi-objective optimization algorithm, MOWAA, by incorporating multi-objective counterparts such as mutation and crossover operations. The algorithm includes innovative strategies like leader selection, a specific archive solution set updating mechanism, and a fixed-size external archive to improve its capacity for solving multi-objective optimization problems.
- 2) Development of a New Two-Archive Mechanism: To achieve simultaneous convergence and diversity, a novel two-archive mechanism is integrated into MOWAA. Unlike conventional single-archive approaches, this strategy separates optimal solutions into two distinct archives with tailored updating schemes, enabling the algorithm to maintain a balance between convergence to the Pareto front and diversity across the objective space.

The structure of the paper is as follows. Section 2 describes the theory and research background of the proposed algorithm MOWAA. Next, experimental results and discussion from numerical benchmark functions, and several engineering problems are presented in Section 3. Following, a discussion on the application of waviness prediction for wire arc additively manufactured (WAAM) steel components is presented in Section 4. Finally, the conclusions and some suggestions for future work are summarized.

## 2. Development of the multi-objective weighted average algorithm

### 2.1. Weighted average algorithm

The authors proposed the weighted average algorithm (WAA), inspired by the weighted average position concept and the Lévy flight, in (Luo et al., 2024). The weighted average position (Hong et al., 2022) refers to a statistical measure that considers the weight or importance of individual elements in a dataset when calculating the overall average position. During the iteration process, the weighted average position replaces the candidate position to explore the feasible search spaces or exploit the approximate solution space. In the exploration process, three different movement strategies in the search space locate the weighted average position, the personal best position, and the global best position as expressed in Eqs. (1)–(3).

$$X_i(t+1) = w_{21} \times (X_{Miu}(t) - X_{PersonalBest}(t)) + w_{22} \times X_{PersonalBest}(t) \quad (1)$$

$$X_i(t+1) = w_{11} \times (X_{Miu}(t) - X_{GlobalBest}(t)) + w_{12} \times (X_{Miu}(t) - X_{PersonalBest}(t)) + w_{13} \times X_{Miu}(t); \quad (2)$$

$$X_i(t+1) = w_{31} \times (X_{Miu}(t) - X_{GlobalBest}(t)) + w_{32} \times X_{GlobalBest}(t); \quad (3)$$

Where  $w_{11}$ ,  $w_{12}$ ,  $w_{13}$ ,  $w_{21}$ ,  $w_{22}$ ,  $w_{31}$ , and  $w_{32}$  are random values ranging from 0 to 1.  $X_{PersonalBest}$ ,  $X_{GlobalBest}$ , and  $X_{Miu}$  are the personal best position for the  $i_{th}$  candidate solution, the global best position for the entire population, and the weighted average position respectively. Besides,  $t$  is

the current iteration.

In the exploitation process, there are two different movement strategies: 1) the Lévy flight (Minh et al., 2022) around the global best position, and 2) replacing old search spaces with new ones, expressed by Eq. (4) and (5) respectively.

$$X_i(t+1) = X_{Global\_Best}(t) + \text{steplength}(S) \quad (4)$$

$$S = \frac{U}{|V|^{1/\beta}}; U = \text{normal}(0, \sigma_u^2); V = \text{normal}(0, \sigma_v^2); \sigma_u = \left\{ \frac{\Gamma(1+\beta)\sin(\pi\beta/2)}{\Gamma[(1+\beta)/2]\beta 2^{(\beta-1)/2}} \right\}^{1/\beta}; \sigma_v = 1; \beta = 1.5 \quad (5)$$

$$X_i(t+1) = \text{rand} \times (UB_{min} - LB_{min}) + LB_{min} \quad (6)$$

$LB_{min}$  and  $UB_{min}$  are defined as the minimum values of lower bound and upper bound in all dimensions.  $S$  is the step length for the Lévy flight.

## 2.2. Genetic algorithm

The genetic algorithm (GA) (D'Angelo and Palmieri, 2021) is a meta-heuristic algorithm inspired by the process of natural selection and genetics, which is used to solve optimization and search problems by mimicking the biological processes of reproduction, mutation, crossover, and selection. In a GA, the problem solution is encoded as a set of strings, which represent individuals in a population. These individuals are evaluated using a fitness function that measures how well they solve the problem. The fittest individuals are then selected to reproduce and create offspring using techniques such as mutation (Eq. (7)) and crossover (Eqs. 8,9). This process is repeated over multiple generations, with the hope that the offspring will inherit the desirable traits of the parents and eventually converge towards an optimal solution.

$$X_i(t+1) = \text{Parent} + \text{rand}(1, \text{numel}(\text{Parent})) \times \text{mutation}_{rate} \quad (7)$$

Where  $\text{Parent}$  is the parent solution position, which is randomly selected from the archive solution set, and  $\text{rand}(1, \text{numel}(\text{Parent}))$  is a random real number from zero to one.

$$\text{crossoverPoint} = \text{randi}([1, \text{numel}(\text{parent1})]); \quad (8)$$

$$X_i(t+1) = [\text{parent1}(1 : \text{crossoverPoint}), \text{parent2}(\text{crossoverPoint} + 1 : \text{end})]; \quad (9)$$

Where  $\text{parent1}$  and  $\text{parent2}$  are the parent solution positions, which are randomly selected from the archive solution set, and  $\text{crossoverPoint}$  is the crossover point for  $\text{parent1}$ .

## 2.3. Multi-objective weighted average algorithm

In this section, the WAA integrated with mutation and crossover as special movement strategies during exploration is proposed to solve multi-objective optimization problems. Multi-objective WAA is based on an external archive to save Pareto-optimal solutions during the optimization process and a crowding distance strategy to preserve the diversity among solutions in this archive. During the iterations, different movement strategies are applied to update the positions of the candidate solutions. Based on a comparison between the current candidate solution's fitness and the corresponding personal best position's fitness, the

outcome determines whether the personal best positions are updated or not. Following, a new archive solution at the  $(i_{th} + 1)$  iteration is generated by applying the non-dominated sorting technique for the entire population's fitness and the archive solution of the  $i_{th}$  iteration. After that, this archive solution will be updated using the non-dominated sorting technique to receive one Pareto-optimal solution within the set size range. As a result, the Pareto-optimal solution obtained in the last iteration will be used to solve the multi-objective

optimization problem. The steps of adapting the WAA towards solving a multi-objective optimization problem are disused in the next subsections.

### 2.3.1. Initialization step

In this phase, a population  $X$  consisting of  $N$  candidate solutions, with each candidate solution defined in  $n$  dimensions, is initialized randomly within the boundary conditions of the problems.

$$X = \begin{bmatrix} x_{1,1} & x_{1,2} & \dots & x_{1,n-1} & x_{1,n} \\ x_{2,1} & x_{2,2} & & x_{2,n-1} & x_{2,n} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ x_{N-1,1} & x_{N-1,2} & \dots & x_{N-1,n-1} & x_{N-1,n} \\ x_{N,1} & x_{N,2} & & x_{N,n-1} & x_{N,n} \end{bmatrix} \quad (10)$$

$$x_{i,j} = \text{rand} \times (UB_j - LB_j) + LB_j, j = 1, 2, \dots, n \quad (11)$$

$UB$  and  $LB$  are the minimum and maximum of the search space. The  $\text{rand}$  function generates random numbers from a standard uniform distribution between 0 and 1. This means that all numbers in this range have equal probability.

Next, a specific function expressed by Eq. (12) is applied to select the suitable search phase (i.e., exploration or exploitation) during the iteration process.

$$f(t) = (\alpha * \text{rand} - 1) * \sin\left(\pi * \frac{t}{\text{Max}_{it}}\right) \quad (12)$$

Where  $t$  is the current iteration,  $\text{Max}_{it}$  is the maximum number of iterations, and  $\alpha$  is a constant to control the balance between exploration and exploitation.

### 2.3.2. Calculation of the weighted average position

The weighted average position of the entire population is calculated in several steps as explained hereafter.

1) Calculation of the maximum and minimum values for each objective.

Assuming that there are  $M$  objective problems and  $N$  candidate solutions  $X_i$ , the fitness of the population for all objective problems can be expressed as:

$$\text{Fitness} = \begin{bmatrix} \text{Fitness}_{1,1} & \text{Fitness}_{1,2} & \dots & \text{Fitness}_{1,M-1} & \text{Fitness}_{1,M} \\ \text{Fitness}_{2,1} & \text{Fitness}_{2,2} & & \text{Fitness}_{2,M-1} & \text{Fitness}_{2,M} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \text{Fitness}_{N-1,1} & \text{Fitness}_{N-1,2} & \dots & \text{Fitness}_{N-1,M-1} & \text{Fitness}_{N-1,M} \\ \text{Fitness}_{N,1} & \text{Fitness}_{N,2} & & \text{Fitness}_{N,M-1} & \text{Fitness}_{N,M} \end{bmatrix} \quad (13)$$

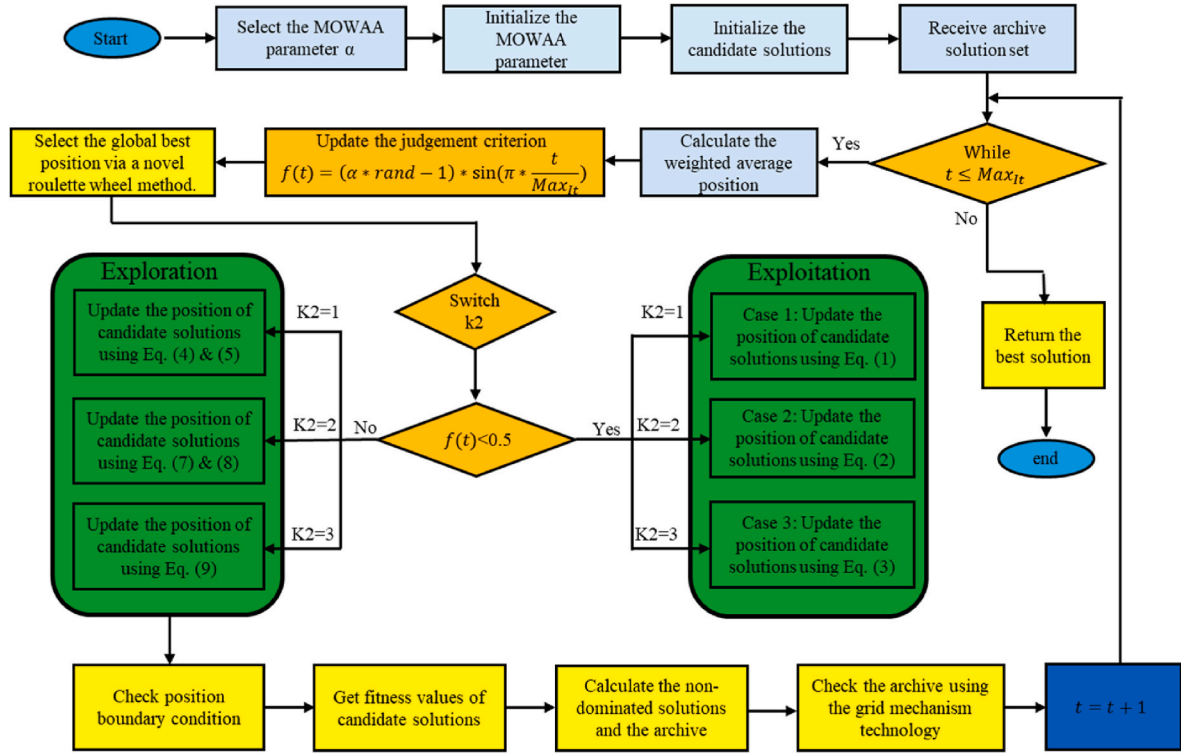


Fig. 1. Flow chart of MOWAA.

Table 1

Results in terms of GD obtained by different MOWAA versions applied to CEC2009 test functions.

| Functions |      | Algorithms |            |            |            |
|-----------|------|------------|------------|------------|------------|
|           |      | MOWAA_M    | MOWAA_C    | MOWAA_O    | MOWAA      |
| UF1       | Ave  | 1.9464E-02 | 1.3183E-02 | 1.5591E-02 | 8.8884E-03 |
|           | Rank | 4          | 2          | 3          | 1          |
|           | Std  | 8.4924E-03 | 5.9250E-03 | 4.1311E-03 | 2.8562E-03 |
| UF2       | Ave  | 7.0366E-03 | 7.1343E-03 | 1.1243E-02 | 6.0284E-03 |
|           | Rank | 2          | 3          | 4          | 1          |
|           | Std  | 1.6292E-03 | 3.7834E-03 | 5.3416E-03 | 1.4413E-03 |
| UF3       | Ave  | 3.2368E-02 | 4.3227E-02 | 3.8385E-02 | 2.4572E-02 |
|           | Rank | 2          | 4          | 3          | 1          |
|           | Std  | 5.1875E-03 | 3.8032E-03 | 3.3656E-03 | 3.2808E-03 |
| UF4       | Ave  | 1.0979E-02 | 1.0458E-02 | 6.1054E-03 | 1.1122E-02 |
|           | Rank | 3          | 2          | 1          | 4          |
|           | Std  | 1.6084E-03 | 1.0057E-03 | 2.6790E-04 | 1.3138E-03 |
| UF5       | Ave  | 2.0668E-01 | 1.2783E-01 | 1.7908E-01 | 1.4813E-01 |
|           | Rank | 4          | 1          | 3          | 2          |
|           | Std  | 3.1034E-02 | 6.4199E-02 | 2.9041E-02 | 5.8416E-02 |
| UF6       | Ave  | 1.0996E-01 | 7.6900E-02 | 8.2423E-02 | 4.5240E-02 |
|           | Rank | 4          | 2          | 3          | 1          |
|           | Std  | 3.4644E-02 | 3.5497E-02 | 2.0459E-02 | 5.0118E-02 |
| UF7       | Ave  | 1.3121E-02 | 1.1044E-02 | 1.3780E-02 | 5.3390E-03 |
|           | Rank | 3          | 2          | 4          | 1          |
|           | Std  | 3.5814E-03 | 9.4922E-03 | 3.0743E-03 | 1.2195E-03 |
| UF8       | Ave  | 1.1487E-02 | 3.4606E-03 | 6.3442E-02 | 2.8304E-03 |
|           | Rank | 3          | 2          | 4          | 1          |
|           | Std  | 5.3223E-03 | 4.7027E-03 | 4.9434E-02 | 2.2543E-03 |
| UF9       | Ave  | 1.4674E-01 | 1.6078E-01 | 1.7770E-01 | 9.4683E-02 |
|           | Rank | 2          | 3          | 4          | 1          |
|           | Std  | 1.2023E-01 | 7.6283E-02 | 7.7610E-02 | 3.7133E-02 |
| UF10      | Ave  | 1.4293E-01 | 3.6127E-03 | 1.3941E-01 | 2.0567E-03 |
|           | Rank | 4          | 2          | 3          | 1          |
|           | Std  | 3.9313E-02 | 7.9875E-03 | 7.6734E-02 | 4.4367E-03 |

Table 2

Results in terms of IGD obtained by different MOWAA versions applied to CEC2009 test functions.

| Functions |      | Algorithms |            |            |            |
|-----------|------|------------|------------|------------|------------|
|           |      | MOWAA_M    | MOWAA_C    | MOWAA_O    | MOWAA      |
| UF1       | Ave  | 4.1487E-03 | 3.3340E-03 | 4.1935E-03 | 3.0314E-03 |
|           | Rank | 3          | 2          | 4          | 1          |
|           | Std  | 6.6727E-04 | 6.9608E-04 | 8.8111E-04 | 2.9123E-04 |
| UF2       | Ave  | 2.8865E-03 | 2.7585E-03 | 2.9181E-03 | 2.5765E-03 |
|           | Rank | 3          | 2          | 4          | 1          |
|           | Std  | 2.1183E-04 | 2.1738E-04 | 2.6852E-04 | 3.4353E-04 |
| UF3       | Ave  | 1.2034E-02 | 1.4158E-02 | 1.2139E-02 | 1.0626E-02 |
|           | Rank | 2          | 4          | 3          | 1          |
|           | Std  | 2.4723E-04 | 5.1868E-04 | 2.8287E-04 | 4.6637E-04 |
| UF4       | Ave  | 3.3069E-03 | 3.1970E-03 | 2.1435E-03 | 3.3379E-03 |
|           | Rank | 3          | 2          | 1          | 4          |
|           | Std  | 3.9014E-04 | 2.7941E-04 | 3.1891E-05 | 3.4428E-04 |
| UF5       | Ave  | 2.8637E-01 | 2.1946E-01 | 3.4284E-01 | 2.3993E-01 |
|           | Rank | 3          | 1          | 4          | 2          |
|           | Std  | 1.6471E-01 | 1.1997E-01 | 5.8148E-02 | 1.2079E-01 |
| UF6       | Ave  | 2.1577E-02 | 1.7603E-02 | 2.1370E-02 | 1.7575E-02 |
|           | Rank | 4          | 2          | 3          | 1          |
|           | Std  | 6.4721E-03 | 4.7427E-03 | 3.2892E-03 | 3.8484E-03 |
| UF7       | Ave  | 3.0140E-03 | 3.6607E-03 | 3.1059E-03 | 2.4766E-03 |
|           | Rank | 2          | 4          | 3          | 1          |
|           | Std  | 2.3704E-04 | 2.3151E-03 | 3.0065E-04 | 1.1102E-04 |
| UF8       | Ave  | 3.3644E-03 | 4.3590E-03 | 3.0873E-03 | 4.1332E-03 |
|           | Rank | 2          | 4          | 1          | 3          |
|           | Std  | 3.2543E-04 | 5.1067E-04 | 2.5778E-04 | 5.1852E-04 |
| UF9       | Ave  | 3.6200E-03 | 4.4499E-03 | 4.7415E-03 | 3.6048E-03 |
|           | Rank | 2          | 3          | 4          | 1          |
|           | Std  | 9.2598E-04 | 6.8107E-04 | 1.1335E-04 | 1.0600E-03 |
| UF10      | Ave  | 7.7845E-03 | 5.6643E-03 | 8.6102E-03 | 5.6290E-03 |
|           | Rank | 3          | 2          | 4          | 1          |
|           | Std  | 1.2426E-03 | 3.3591E-04 | 1.4419E-03 | 1.2747E-04 |

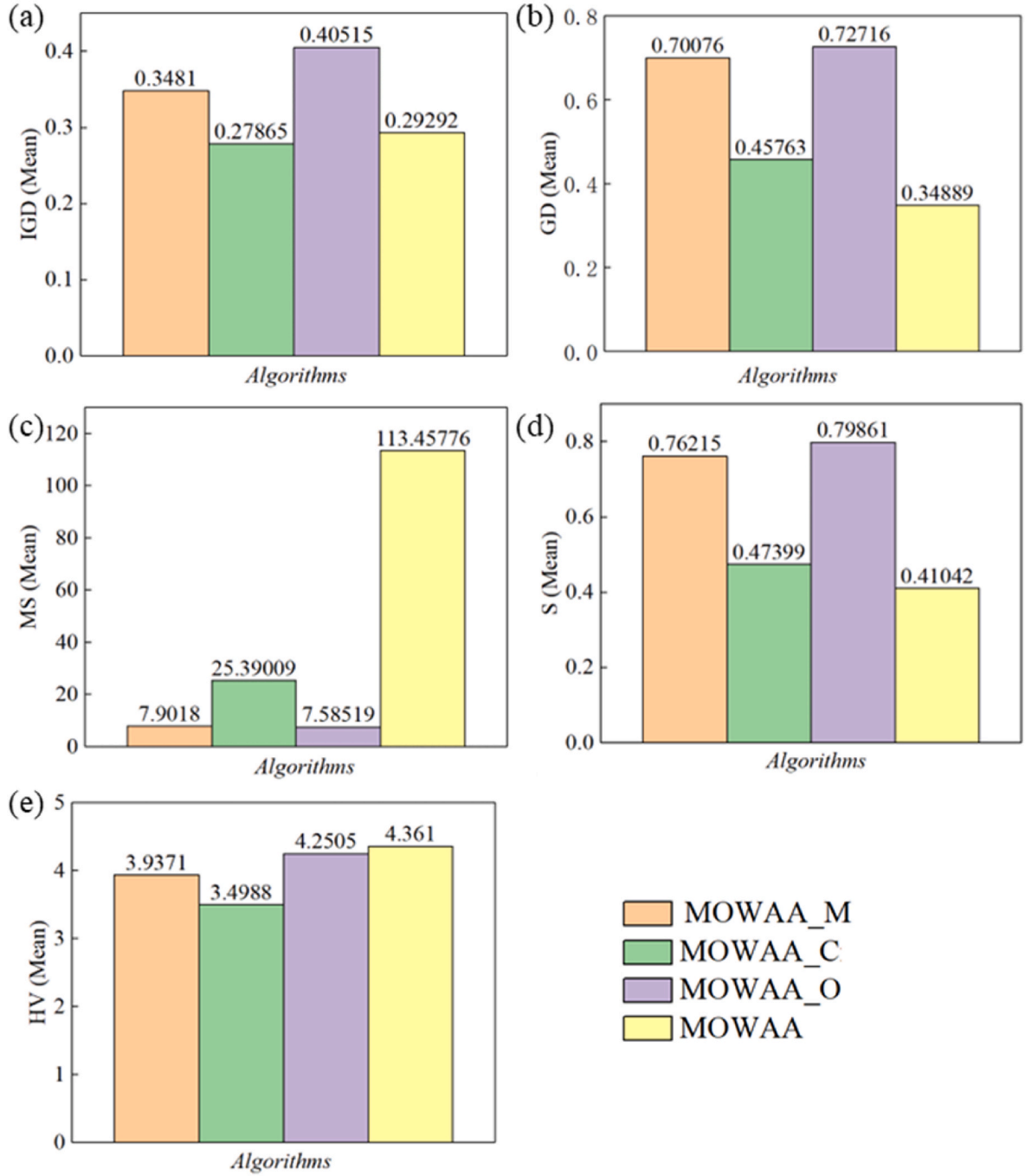


Fig. 2. Average value of different metrics for all CEC2009 benchmark test functions: (a) IGD; (b) GD; (c) MS; (d) S; (e) HV.

$$Max = \max(Fitness(:, 2)); Min = \min(Fitness(:, 2)); \quad (14)$$

Where  $Fitness_{ij}$  stands for the fitness of the  $i_{th}$  candidate solution for the  $j_{th}$  objective function. Max and Min are the maximum and minimum values for the M objective problems respectively. Besides,  $\max$  and  $\min$  functions return the maximum and minimum values, respectively, along a specified dimension of an array.

## 2) Normalization of fitness values

$$Normalized_{Fitness}(i, j) = \frac{Fitness(i, j) - Min(1, j)}{Max(1, j) - Min(1, j)} \quad (15)$$

Where  $Fitness(i, j)$  are the fitness values of the  $i_{th}$  candidate solution and the  $j_{th}$  objective function,  $Min(1, j)$  and  $Max(1, j)$  are the maximum

and minimum values for the  $j_{th}$  objective function.  $Normalized_{Fitness}(i, j)$  are the normalized fitness values of the  $i_{th}$  candidate solution and the  $j_{th}$  objective function. ( $j = 1 \sim M, i = 1 \sim N$ )

## 3) Summation of the normalized fitness values for each candidate solution

$$Sum_{Fitness}(i) = \sum(Normalized_{Fitness}, 2); i = 1 \sim N \quad (16)$$

$$Total_{Fitness}(i) = \sum(Sum_{Fitness}); \quad (17)$$

Where  $Sum_{Fitness}(i)$  is the total fitness of all objective functions for the  $i_{th}$  candidate solution, and  $Total_{Fitness}$  is the total fitness for the entire population.

**Table 3**

Results in terms of MS obtained by different MOWAA versions applied to CEC2009 test functions.

| Functions |      | Algorithms |            |            |            |
|-----------|------|------------|------------|------------|------------|
|           |      | MOWAA_M    | MOWAA_C    | MOWAA_O    | MOWAA      |
| UF1       | Ave  | 7.7200E-01 | 8.3094E-01 | 7.5326E-01 | 9.9768E-01 |
|           | Rank | 3          | 2          | 4          | 1          |
|           | Std  | 1.7678E-01 | 9.3684E-02 | 7.1360E-02 | 1.2896E-01 |
| UF2       | Ave  | 9.3832E-01 | 1.0119E+00 | 8.9290E-01 | 9.1033E-01 |
|           | Rank | 2          | 1          | 4          | 3          |
|           | SD   | 5.6847E-02 | 1.4505E-01 | 1.0140E-01 | 5.0507E-02 |
| UF3       | Ave  | 9.9701E-01 | 1.6107E+00 | 1.0005E+00 | 9.9205E-01 |
|           | Rank | 3          | 1          | 2          | 4          |
|           | SD   | 8.9811E-03 | 3.6959E-01 | 4.5600E-04 | 2.3855E-02 |
| UF4       | Ave  | 9.9354E-01 | 9.7019E-01 | 9.9948E-01 | 9.9101E-01 |
|           | Rank | 2          | 4          | 1          | 3          |
|           | SD   | 1.8873E-02 | 2.6532E-02 | 2.8698E-03 | 1.2858E-02 |
| UF5       | Ave  | 3.6279E-01 | 8.1177E-01 | 3.3961E-01 | 8.1157E+01 |
|           | Rank | 3          | 2          | 4          | 1          |
|           | SD   | 8.4535E-02 | 2.7826E-01 | 8.1895E-02 | 2.4004E+02 |
| UF6       | Ave  | 5.4483E-01 | 9.3758E-01 | 5.4843E-01 | 9.4558E-01 |
|           | Rank | 4          | 2          | 3          | 1          |
|           | SD   | 2.9283E-01 | 6.4260E-01 | 1.9204E-01 | 5.0473E-01 |
| UF7       | Ave  | 8.1623E-01 | 9.1412E-01 | 7.7210E-01 | 9.2546E-01 |
|           | Rank | 3          | 2          | 4          | 1          |
|           | SD   | 7.3786E-02 | 1.3900E-01 | 9.0101E-02 | 4.1692E-02 |
| UF8       | Ave  | 9.9514E-01 | 1.8094E+00 | 7.5835E-01 | 1.9388E+00 |
|           | Rank | 3          | 2          | 4          | 1          |
|           | SD   | 1.6245E-01 | 7.8423E-01 | 5.2508E-02 | 1.2553E+00 |
| UF9       | Ave  | 6.9079E-01 | 7.2808E-01 | 7.2398E-01 | 8.2670E-01 |
|           | Rank | 4          | 2          | 3          | 1          |
|           | SD   | 7.5187E-02 | 1.0751E-01 | 7.7606E-02 | 1.4469E-01 |
| UF10      | Ave  | 7.9115E-01 | 1.5765E+01 | 7.9658E-01 | 2.3773E+01 |
|           | Rank | 4          | 2          | 3          | 1          |
|           | SD   | 6.9620E-02 | 1.3749E+01 | 9.5822E-02 | 1.5025E+01 |

**Table 4**

Results in terms of S obtained by different MOWAA versions applied to CEC2009 test functions.

| Functions |      | Algorithms |            |            |            |
|-----------|------|------------|------------|------------|------------|
|           |      | MOWAA_M    | MOWAA_C    | MOWAA_O    | MOWAA      |
| UF1       | Ave  | 2.0959E-02 | 2.2951E-02 | 1.7077E-02 | 1.4307E-02 |
|           | Rank | 3          | 4          | 2          | 1          |
|           | Std  | 5.7624E-03 | 1.0925E-02 | 3.7574E-03 | 4.3633E-03 |
| UF2       | Ave  | 1.7747E-02 | 1.6557E-02 | 1.8188E-02 | 1.5751E-02 |
|           | Rank | 3          | 2          | 4          | 1          |
|           | Std  | 6.1706E-03 | 4.7402E-03 | 6.1480E-03 | 3.3803E-03 |
| UF3       | Ave  | 1.4581E-02 | 2.1481E-02 | 1.0538E-02 | 2.0620E-02 |
|           | Rank | 2          | 4          | 1          | 3          |
|           | Std  | 3.5177E-03 | 1.4685E-02 | 4.4404E-03 | 7.2224E-03 |
| UF4       | Ave  | 1.0974E-02 | 1.2338E-02 | 1.0850E-02 | 1.2513E-02 |
|           | Rank | 2          | 3          | 1          | 4          |
|           | Std  | 2.2116E-03 | 2.5426E-03 | 3.6331E-03 | 1.9390E-03 |
| UF5       | Ave  | 8.2606E-02 | 5.5622E-02 | 9.2627E-02 | 5.4894E-02 |
|           | Rank | 3          | 2          | 4          | 1          |
|           | Std  | 2.6405E-02 | 2.8829E-02 | 3.9913E-02 | 2.4555E-02 |
| UF6       | Ave  | 1.0454E-01 | 4.5315E-02 | 3.9841E-02 | 4.3304E-02 |
|           | Rank | 4          | 3          | 1          | 2          |
|           | Std  | 1.0034E-01 | 3.0089E-02 | 1.3316E-02 | 2.9389E-02 |
| UF7       | Ave  | 2.0766E-02 | 3.7880E-02 | 3.0355E-02 | 1.6518E-02 |
|           | Rank | 2          | 4          | 3          | 1          |
|           | Std  | 4.7812E-03 | 6.0376E-02 | 2.2037E-02 | 3.6360E-03 |
| UF8       | Ave  | 7.8338E-02 | 4.5159E-02 | 1.4033E-01 | 6.2643E-02 |
|           | Rank | 3          | 1          | 4          | 2          |
|           | Std  | 1.4115E-02 | 1.5948E-02 | 6.1894E-02 | 2.8851E-02 |
| UF9       | Ave  | 1.4058E-01 | 1.8212E-01 | 1.9204E-01 | 1.3955E-01 |
|           | Rank | 2          | 3          | 4          | 1          |
|           | Std  | 7.4436E-02 | 5.4473E-02 | 6.8215E-02 | 3.0956E-02 |
| UF10      | Ave  | 2.7106E-01 | 3.4573E-02 | 2.4676E-01 | 3.0323E-02 |
|           | Rank | 4          | 2          | 3          | 1          |
|           | Std  | 1.1187E-01 | 1.2401E-02 | 8.8154E-02 | 1.4199E-02 |

**Table 5**

Results in terms of HV obtained by different MOWAA versions applied to CEC2009 test functions.

| Functions |      | Algorithms |            |            |            |
|-----------|------|------------|------------|------------|------------|
|           |      | MOWAA_M    | MOWAA_C    | MOWAA_O    | MOWAA      |
| UF1       | Ave  | 5.1100E-01 | 5.5600E-01 | 6.0300E-01 | 6.4100E-01 |
|           | Rank | 4          | 3          | 2          | 1          |
|           | Std  | 9.8712E-02 | 6.4257E-02 | 5.1561E-02 | 6.6685E-02 |
| UF2       | Ave  | 6.2990E-01 | 6.0650E-01 | 6.2300E-01 | 6.5700E-01 |
|           | Rank | 2          | 4          | 3          | 1          |
|           | Std  | 4.7608E-02 | 7.5064E-02 | 5.9324E-02 | 4.6349E-02 |
| UF3       | Ave  | 1.3410E-01 | 1.9100E-01 | 1.1160E-01 | 2.0030E-01 |
|           | Rank | 3          | 2          | 4          | 1          |
|           | Std  | 1.4714E-02 | 7.8758E-02 | 1.2435E-02 | 2.8468E-02 |
| UF4       | Ave  | 2.7120E-01 | 2.8460E-01 | 2.6310E-01 | 3.0500E-01 |
|           | Rank | 3          | 2          | 4          | 1          |
|           | Std  | 2.0024E-02 | 2.9104E-02 | 9.6794E-03 | 1.8300E-02 |
| UF5       | Ave  | 3.6510E-01 | 1.8800E-01 | 4.9190E-01 | 2.5800E-01 |
|           | Rank | 2          | 4          | 1          | 3          |
|           | Std  | 8.3751E-02 | 1.2600E-01 | 9.2054E-02 | 1.4280E-01 |
| UF6       | Ave  | 5.2800E-01 | 2.9290E-01 | 5.0450E-01 | 5.4780E-01 |
|           | Rank | 2          | 4          | 3          | 1          |
|           | Std  | 1.8985E-01 | 2.1733E-01 | 1.7649E-01 | 2.0416E-01 |
| UF7       | Ave  | 4.5900E-01 | 4.3350E-01 | 4.4400E-01 | 4.8010E-01 |
|           | Rank | 2          | 4          | 3          | 1          |
|           | Std  | 8.5269E-02 | 1.2685E-01 | 9.9470E-02 | 5.6684E-02 |
| UF8       | Ave  | 3.2500E-01 | 2.0030E-01 | 5.0420E-01 | 1.9510E-01 |
|           | Rank | 2          | 3          | 1          | 4          |
|           | Std  | 1.0340E-01 | 1.5982E-02 | 9.2937E-02 | 9.1591E-03 |
| UF9       | Ave  | 4.5800E-01 | 5.3290E-01 | 4.7550E-01 | 5.5270E-01 |
|           | Rank | 4          | 2          | 3          | 1          |
|           | Std  | 1.1262E-01 | 8.8959E-02 | 6.4582E-02 | 1.3960E-01 |
| UF10      | Ave  | 2.5580E-01 | 2.1310E-01 | 2.2970E-01 | 5.2400E-01 |
|           | Rank | 2          | 4          | 3          | 1          |
|           | Std  | 8.2490E-02 | 8.5765E-02 | 1.1476E-01 | 1.1216E-01 |

#### 4) Calculation of the weighted average position

$$X_{Mit} = \frac{\sum_{i=1}^N X_i \times (Total_{Fitness} - Sum_{Fitness}(X_i))}{Total_{Fitness} \times (N - 1)} \quad (18)$$

Where  $X_{Mit}$  is the weighted average position for the entire population.

#### 2.3.3. The leader selection mechanism

Traditionally, the leader selection plays a crucial role in guiding the other candidate solutions towards promising regions of the search space with the aim of achieving a solution close to the global optimum. However, in multi-objective optimization problems, it is very difficult to perform the leader selection through the fitness value. The leader selection step then chooses one of these non-dominated solutions and strategically places it in the least congested region of the search space. This innovative approach helps address the complexities associated with Pareto optimality in a multi-objective optimization scenario. In the current research, a roulette wheel selection method (Ding et al., 2023) is used to select the global best position:

$$P_i = \frac{C}{M_i} \quad (19)$$

Where  $M_i$  is the variety of obtained Pareto-optimal solutions in the  $i_{th}$  section and  $C$  is a constant with value greater than one. Based on this equation, hypercubes with less congestion are more likely to be selected as the new leader. Besides, the use of a roulette wheel provides a low chance to select a new leader from the most populated hypercubes.

#### 2.3.4. The archive iteration strategy

A fixed-sized external archive is integrated into the WAA for saving the obtained non-dominated Pareto-optimal solution. archive.

In the  $(i_{th} + 1)$  iteration, a set is generated and consists of the candidate solutions' fitness, the personal best positions' fitness and the

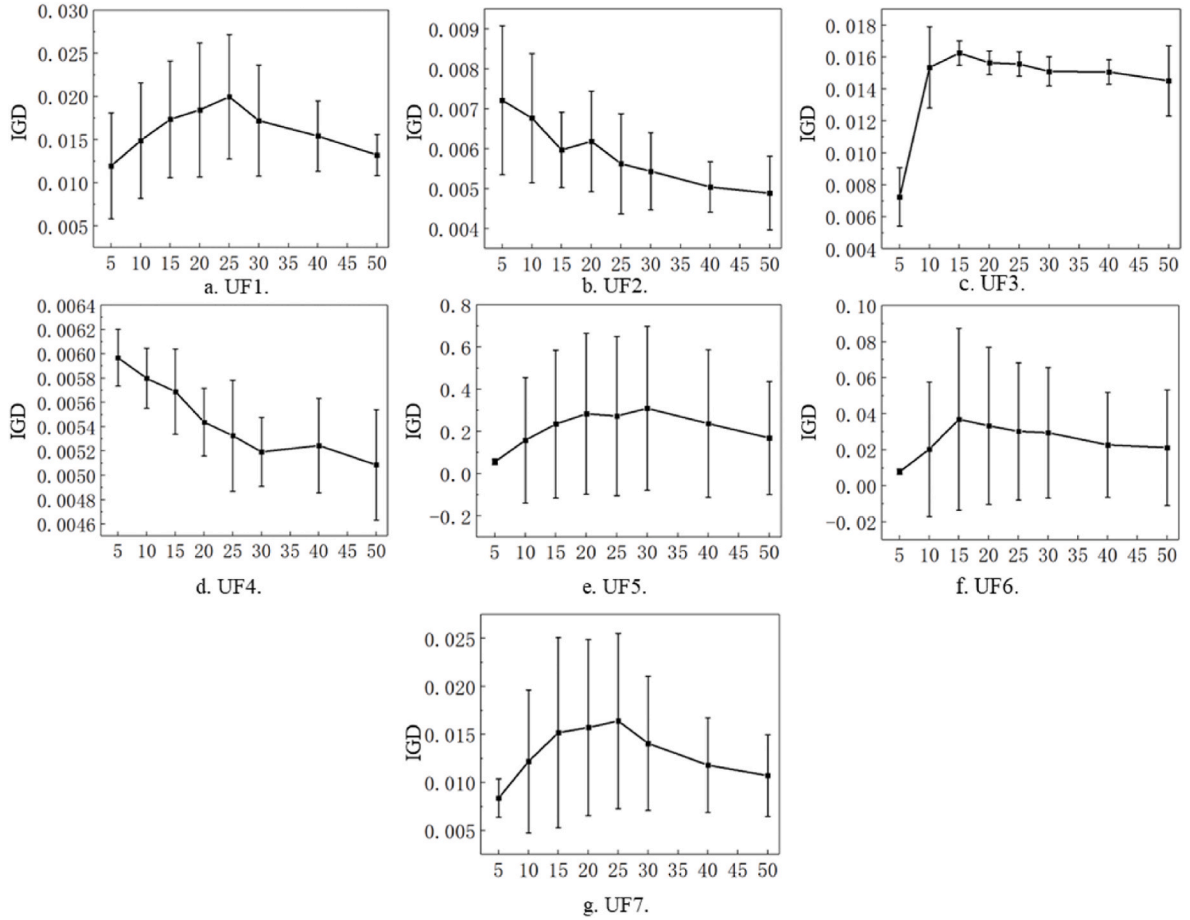


Fig. 3. IGD values for different numbers of iterations applied to functions UF1–UF7.

archive solution set of the previous  $i_{th}$  iteration. After that, the non-dominated sorting technique is applied to select the non-dominated solutions from the above generated set. The selected non-dominated solutions are regarded as the archive solution set in the  $(i_{th} + 1)$  iteration times.

### 2.3.5. Complete MOWAA algorithm

This subsection combines the steps described in the previous subsections to build the complete MOWAA. In the initial stage of the MOWAA, a random set of candidate solutions is generated in the search space. After that, during the iteration process, several movement strategies are applied to keep a balance between exploration and exploitation resulting in the variation of the population. Once reaching a termination criterion, the archive solution set is regarded as the Pareto-optimal solutions. The pseudo-code of MOWAA is shown in Algorithm 1, and the flowchart of the WAA can be seen in Fig. 1.

#### Algorithm 1. Pseudo-code of the MOWAA algorithm

1. Initialization phase:
2. Initialize the solutions candidates' positions randomly. (Solutions:  $i = 1, \dots, N$ .)
3. Calculate the fitness function values.
4. Receive archive solution set: determine the obtained Pareto-optimal solution.
5. WHILE (The end condition is not met) do
6. Calculate the weighted average position of the current population.
7. for ( $i = 1, 2, \dots, N$ ) do
8.  $f(t) = (\alpha * rand - 1) * \sin\left(\pi * \frac{t}{Max_{it}}\right)$ ; %judge criterion: exploration or exploitation
9.  $k2 = randi(3)$ ; % select random the first, the second, the third movement strategy.
10. Select the global best position via a novel roulette wheel method.

(continued on next column)

(continued)

11. If  $k1 < 0.5$
12. Switch  $k2$
13. Case 1 % the first movement strategy for exploration
14. Update the current solution using Eq. (1).
15. Case 2 % the second movement strategy for exploration
16. Update the current solution using Eq. (2).
17. Case 3 % the third movement strategy for exploration
18. Update the current solution using Eq. (3).
19. End
20. Else
21. Switch  $k2$
22. Case 1 % the first movement strategy for exploitation
23. Update the current solution using Eq. (4) and Eq. (5).
24. Case 2 % the first movement strategy for exploitation
25. Update the current solution using Eq. (7) and Eq. (8).
26. Case 3 % the first movement strategy for exploitation
27. Update the current solution using Eq. (9).
28. End
29. End
30. Check position bounds
31. Get fitness values of candidate solutions.
32. End for
33. Calculate the non-dominated solutions and the archive.
34. Check the archive using the grid mechanism technology.
35. End while
36. Return the archive solution set.

### 3. Results and discussion

This section presents a comprehensive evaluation of the proposed MOWAA algorithm on both two-objective and three-objective

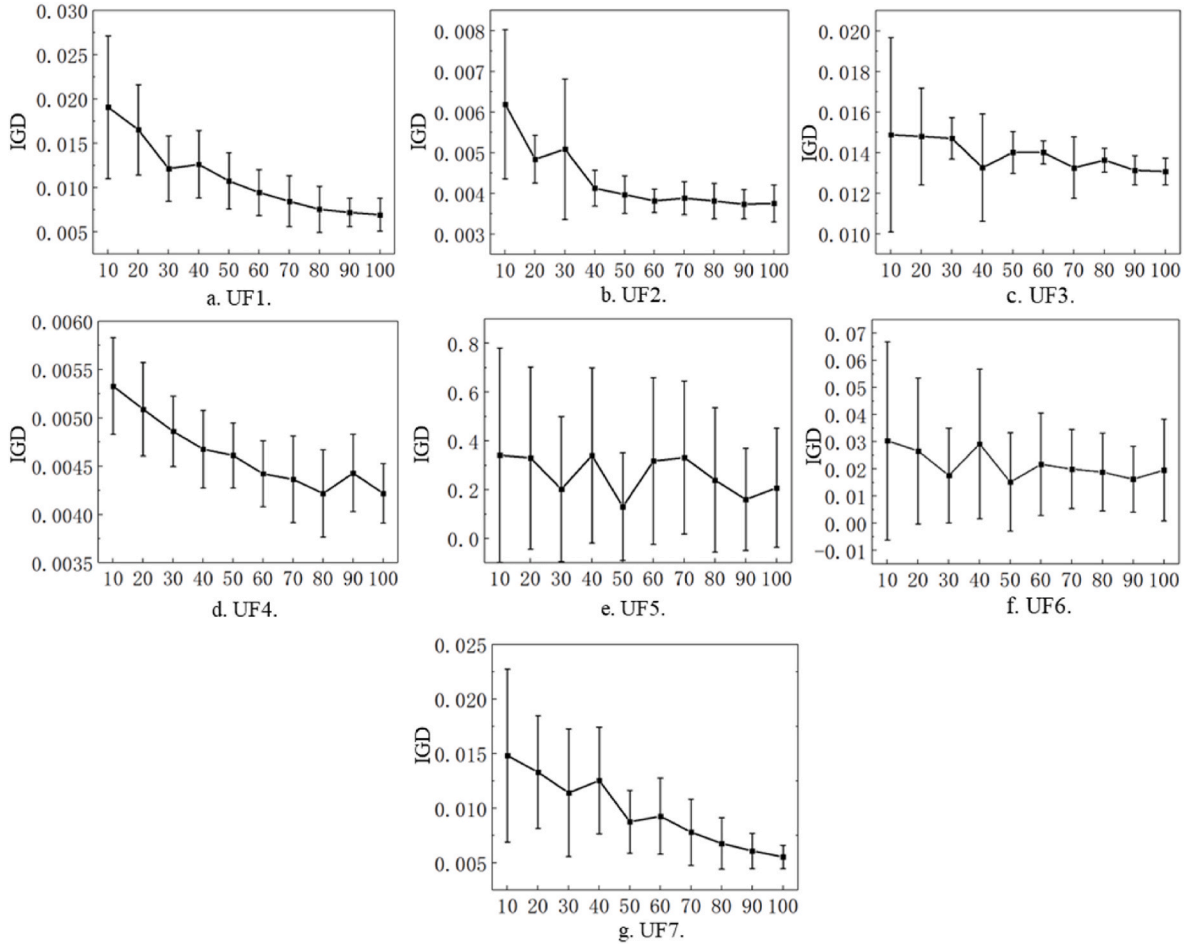


Fig. 4. IGD values for different values of population size applied to functions UF1–UF7.

Table 6

Parameter values for the different algorithms.

| Parameters                     | MOBSLO | MOGOA | MOAVOA | MOWAA |
|--------------------------------|--------|-------|--------|-------|
| Mutation rate                  | –      | –     | –      | 0.1   |
| Control parameter ( $\alpha$ ) | –      | –     | –      | 10    |
| Maximum number of iterations   | 3000   | 3000  | 3000   | 3000  |
| Population size                | 100    | 100   | 100    | 100   |
| Archive size                   | 100    | 100   | 100    | 100   |
| Number of adaptive grids       | 10     | 10    | 10     | 10    |
| Grid inflation parameter       | 0.1    | 0.1   | 0.1    | 0.1   |
| Beta                           | 4      | 4     | 4      | 4     |
| Gamma                          | 2      | 2     | 2      | 2     |

benchmark functions. All experiments are conducted on a Windows 10 (64-bit) system with an Intel Core i7-8750H processor (2.20 GHz), 8 GB RAM, using MATLAB 2013b.

The section is organized as follows.

- **Subsection 3.1 (Performance Metrics):** Introduces the evaluation criteria for multi-objective optimization algorithms.
- **Subsection 3.2 (Benchmark Functions):** Details of the test functions used for experimental validation.
- **Subsection 3.3 (Component Analysis):** Examines the individual contribution of each algorithmic component to MOWAA's performance.
- **Subsection 3.4 (Parameter Sensitivity):** Investigates the impact of key parameters on the algorithm's effectiveness.

- **Subsection 3.5 (Comparative Study):** Benchmarks MOWAA against state-of-the-art multi-objective optimization algorithms.

### 3.1. Performance metrics

Different metrics are used to evaluate the algorithm performance.

- 1) The IGD and GD are two widely used metrics for assessing the convergence of multi-objective meta-heuristic optimization algorithms. Regarding GD, it is computed as the average distance from each solution in the obtained set to the nearest point on the true Pareto front, as expressed in Eq. (20). On the contrary, IGD calculates the average distance from each point on the true Pareto front to the nearest point in the obtained set, as expressed by Eq. (21). Thus, smaller values of GD and IGD indicate better performance, implying that the generated solutions are closer to the true Pareto front.

$$IGD(P^*, \Omega) = \frac{\sum_{x \in P^*} \text{dis}(x, \Omega)}{|P^*|} \quad (20)$$

$$GD = \left( \frac{1}{n_{pf}} \sum_{i=1}^{n_{pf}} \text{dis}_i^2 \right)^{\frac{1}{2}} \quad (21)$$

Where  $n_{pf}$  is the number of points in the obtained Pareto front;  $\text{dis}_i^2$  is the minimum distance between the obtained Pareto front and the true Pareto front;  $P^*$  is the set of points on the true Pareto front;  $\Omega$  is the non-dominated solution set;  $\text{dis}(x, \Omega)$  is the minimum Euclidean distance

**Table 7**  
Results in terms of GD obtained by different algorithms applied to CEC2009 test functions.

| Functions |      | Algorithms |          |          |          |
|-----------|------|------------|----------|----------|----------|
|           |      | MOWAA      | MOBSLO   | MOAVOA   | MOGOA    |
| UF1       | Ave  | 8.89E-03   | 4.93E-03 | 2.40E-02 | 1.82E-02 |
|           | Rank | 2          | 1        | 4        | 3        |
|           | Std  | 2.86E-03   | 2.41E-03 | 8.26E-03 | 5.84E-03 |
| UF2       | Ave  | 6.03E-03   | 7.21E-03 | 8.00E-03 | 6.06E-03 |
|           | Rank | 1          | 3        | 4        | 2        |
|           | Std  | 1.44E-03   | 5.01E-03 | 4.24E-03 | 2.62E-03 |
| UF3       | Ave  | 2.46E-02   | 2.69E-02 | 2.04E-02 | 2.30E-02 |
|           | Rank | 3          | 4        | 1        | 2        |
|           | Std  | 3.28E-03   | 8.78E-03 | 4.86E-03 | 3.43E-03 |
| UF4       | Ave  | 1.11E-02   | 6.66E-03 | 6.56E-03 | 9.84E-03 |
|           | Rank | 4          | 2        | 1        | 3        |
|           | Std  | 1.31E-03   | 5.16E-04 | 5.80E-04 | 8.28E-05 |
| UF5       | Ave  | 1.48E-01   | 6.82E-02 | 2.62E-01 | 1.50E-01 |
|           | Rank | 2          | 1        | 4        | 3        |
|           | Std  | 5.84E-02   | 5.58E-02 | 6.44E-02 | 2.92E-02 |
| UF6       | Ave  | 4.52E-02   | 3.75E-02 | 2.02E-01 | 6.47E-02 |
|           | Rank | 2          | 1        | 4        | 3        |
|           | Std  | 5.01E-02   | 2.18E-02 | 9.13E-02 | 1.49E-02 |
| UF7       | Ave  | 5.34E-03   | 5.43E-03 | 2.23E-02 | 1.46E-02 |
|           | Rank | 1          | 2        | 4        | 3        |
|           | Std  | 1.22E-03   | 2.42E-03 | 9.53E-03 | 1.04E-02 |
| UF8       | Ave  | 2.83E-03   | 9.78E-02 | 1.21E-01 | 1.29E-01 |
|           | Rank | 1          | 2        | 3        | 4        |
|           | Std  | 2.25E-03   | 5.64E-02 | 6.45E-02 | 1.43E-02 |
| UF9       | Ave  | 9.47E-02   | 2.00E-01 | 3.00E-01 | 1.08E-01 |
|           | Rank | 1          | 3        | 4        | 2        |
|           | Std  | 3.71E-02   | 1.02E-01 | 9.65E-02 | 4.24E-02 |
| UF10      | Ave  | 2.06E-03   | 5.03E-01 | 2.41E-01 | 4.43E-01 |
|           | Rank | 1          | 4        | 2        | 3        |
|           | Std  | 4.44E-03   | 2.24E-01 | 1.36E-01 | 5.47E-02 |

**Table 8**  
Results in terms of IGD obtained by different algorithms applied to CEC2009 test functions.

| Functions |      | Algorithms |          |          |          |
|-----------|------|------------|----------|----------|----------|
|           |      | MOWAA      | MOBSLO   | MOAVOA   | MOGOA    |
| UF1       | Ave  | 3.03E-03   | 3.24E-03 | 3.46E-03 | 8.63E-03 |
|           | Rank | 1          | 2        | 3        | 4        |
|           | Std  | 2.91E-04   | 4.03E-04 | 7.17E-04 | 1.90E-03 |
| UF2       | Ave  | 2.58E-03   | 2.01E-03 | 2.19E-03 | 3.53E-03 |
|           | Rank | 3          | 1        | 2        | 4        |
|           | Std  | 3.44E-04   | 3.18E-04 | 3.07E-04 | 5.34E-04 |
| UF3       | Ave  | 1.06E-02   | 9.29E-03 | 8.13E-03 | 9.26E-03 |
|           | Rank | 4          | 3        | 1        | 2        |
|           | Std  | 4.66E-04   | 2.81E-03 | 6.82E-04 | 7.45E-04 |
| UF4       | Ave  | 3.34E-03   | 2.28E-03 | 2.26E-03 | 3.24E-03 |
|           | Rank | 4          | 2        | 1        | 3        |
|           | Std  | 3.44E-04   | 8.41E-05 | 7.34E-05 | 4.83E-05 |
| UF5       | Ave  | 2.40E-01   | 8.54E-02 | 3.57E-01 | 3.16E-01 |
|           | Rank | 2          | 1        | 4        | 3        |
|           | Std  | 1.21E-01   | 4.55E-02 | 7.64E-02 | 3.79E-02 |
| UF6       | Ave  | 1.76E-02   | 8.80E-03 | 2.99E-02 | 2.09E-02 |
|           | Rank | 2          | 1        | 4        | 3        |
|           | Std  | 3.85E-03   | 3.48E-03 | 1.20E-02 | 5.96E-03 |
| UF7       | Ave  | 2.48E-03   | 2.81E-03 | 2.93E-03 | 1.09E-02 |
|           | Rank | 1          | 2        | 3        | 4        |
|           | Std  | 1.11E-04   | 2.75E-03 | 5.64E-04 | 8.63E-03 |
| UF8       | Ave  | 4.13E-03   | 4.78E-03 | 3.08E-03 | 5.90E-03 |
|           | Rank | 2          | 3        | 1        | 4        |
|           | Std  | 5.19E-04   | 1.14E-03 | 4.46E-04 | 1.90E-04 |
| UF9       | Ave  | 3.60E-03   | 3.51E-03 | 4.85E-03 | 6.61E-03 |
|           | Rank | 2          | 1        | 3        | 4        |
|           | Std  | 1.06E-03   | 8.13E-04 | 9.85E-04 | 1.14E-03 |
| UF10      | Ave  | 5.63E-03   | 7.72E-03 | 4.44E-03 | 3.03E-02 |
|           | Rank | 2          | 3        | 1        | 4        |
|           | Std  | 1.27E-04   | 1.57E-03 | 1.21E-03 | 2.25E-03 |

**Table 9**  
Results in terms of MS obtained by different algorithms applied to CEC2009 test functions.

| Functions |      | Algorithms |          |          |          |
|-----------|------|------------|----------|----------|----------|
|           |      | MOWAA      | MOBSLO   | MOAVOA   | MOGOA    |
| UF1       | Ave  | 9.98E-01   | 8.22E-01 | 1.49E+00 | 7.27E-01 |
|           | Rank | 2          | 3        | 1        | 4        |
|           | Std  | 1.29E-01   | 1.37E-01 | 1.75E-01 | 1.51E-01 |
| UF2       | Ave  | 9.10E-01   | 1.10E+00 | 1.13E+00 | 8.85E-01 |
|           | Rank | 3          | 2        | 1        | 4        |
|           | Std  | 5.05E-02   | 1.28E-01 | 1.09E-01 | 3.53E-02 |
| UF3       | Ave  | 9.92E-01   | 9.68E-01 | 1.00E+00 | 6.05E-01 |
|           | Rank | 2          | 3        | 1        | 4        |
|           | Std  | 2.39E-02   | 1.99E-01 | 0.00E+00 | 1.10E-01 |
| UF4       | Ave  | 9.91E-01   | 9.92E-01 | 9.94E-01 | 9.05E-01 |
|           | Rank | 3          | 2        | 1        | 4        |
|           | Std  | 1.29E-02   | 9.80E-03 | 1.11E-02 | 1.39E-02 |
| UF5       | Ave  | 8.12E+01   | 1.67E+00 | 2.90E+00 | 2.38E-01 |
|           | Rank | 1          | 3        | 2        | 4        |
|           | Std  | 2.40E+02   | 6.04E-01 | 4.35E-01 | 1.13E-01 |
| UF6       | Ave  | 9.46E-01   | 1.31E+00 | 3.17E+00 | 2.53E-01 |
|           | Rank | 3          | 2        | 1        | 4        |
|           | Std  | 5.05E-01   | 5.69E-01 | 9.09E-01 | 1.29E-01 |
| UF7       | Ave  | 9.25E-01   | 1.04E+00 | 1.46E+00 | 8.46E-01 |
|           | Rank | 3          | 2        | 1        | 4        |
|           | Std  | 4.17E-02   | 1.75E-01 | 1.99E-01 | 7.92E-02 |
| UF8       | Ave  | 1.94E+00   | 1.84E+00 | 2.61E+00 | 4.42E-01 |
|           | Rank | 2          | 3        | 1        | 4        |
|           | Std  | 1.26E+00   | 7.71E-01 | 1.21E+00 | 1.59E-01 |
| UF9       | Ave  | 8.27E-01   | 4.57E+00 | 5.88E+00 | 1.94E-01 |
|           | Rank | 3          | 2        | 1        | 4        |
|           | Std  | 1.45E-01   | 1.68E+00 | 1.42E+00 | 7.30E-02 |
| UF10      | Ave  | 2.38E+01   | 6.01E+00 | 3.67E+00 | 3.23E-01 |
|           | Rank | 1          | 2        | 3        | 4        |
|           | Std  | 1.50E+01   | 2.19E+00 | 1.57E+00 | 1.24E-01 |

**Table 10**  
Results in terms of S obtained by different algorithms applied to CEC2009 test functions.

| Functions |      | Algorithms |          |          |          |
|-----------|------|------------|----------|----------|----------|
|           |      | MOWAA      | MOBSLO   | MOAVOA   | MOGOA    |
| UF1       | Ave  | 1.43E-04   | 6.18E-03 | 1.38E-02 | 1.73E-04 |
|           | Rank | 1          | 3        | 4        | 2        |
|           | Std  | 4.36E-05   | 4.08E-03 | 4.86E-03 | 8.60E-05 |
| UF2       | Ave  | 1.58E-02   | 1.02E-02 | 1.05E-02 | 8.96E-02 |
|           | Rank | 3          | 1        | 2        | 4        |
|           | Std  | 3.38E-03   | 3.43E-03 | 3.24E-03 | 3.85E-02 |
| UF3       | Ave  | 2.06E-02   | 9.80E-03 | 7.30E-03 | 4.73E-02 |
|           | Rank | 3          | 2        | 1        | 4        |
|           | Std  | 7.22E-03   | 4.31E-03 | 2.08E-03 | 1.23E-04 |
| UF4       | Ave  | 1.25E-03   | 9.94E-03 | 1.09E-02 | 4.33E-03 |
|           | Rank | 1          | 3        | 4        | 2        |
|           | Std  | 1.94E-03   | 2.08E-03 | 3.30E-03 | 4.77E-03 |
| UF5       | Ave  | 5.49E-03   | 4.62E-02 | 9.59E-02 | 6.51E-03 |
|           | Rank | 1          | 3        | 4        | 2        |
|           | Std  | 2.46E-03   | 5.24E-02 | 3.47E-02 | 2.59E-03 |
| UF6       | Ave  | 4.33E-02   | 2.78E-02 | 6.37E-02 | 3.71E-02 |
|           | Rank | 3          | 1        | 4        | 2        |
|           | Std  | 2.94E-02   | 3.32E-02 | 3.44E-02 | 9.06E-02 |
| UF7       | Ave  | 1.65E-02   | 1.01E-02 | 1.57E-02 | 4.83E-02 |
|           | Rank | 3          | 1        | 2        | 4        |
|           | Std  | 3.64E-03   | 3.42E-03 | 6.62E-03 | 3.61E-02 |
| UF8       | Ave  | 6.26E-03   | 5.89E-02 | 1.02E-01 | 8.80E-03 |
|           | Rank | 1          | 3        | 4        | 2        |
|           | Std  | 2.89E-02   | 2.07E-02 | 2.82E-02 | 1.12E-02 |
| UF9       | Ave  | 1.40E-03   | 1.40E-01 | 1.60E-01 | 1.91E-02 |
|           | Rank | 1          | 3        | 4        | 2        |
|           | Std  | 3.10E-03   | 9.95E-02 | 5.29E-02 | 1.71E-02 |
| UF10      | Ave  | 3.03E-02   | 1.64E-01 | 1.44E-01 | 1.31E-01 |
|           | Rank | 1          | 4        | 3        | 2        |
|           | Std  | 1.42E-02   | 1.18E-01 | 6.86E-02 | 1.99E-02 |

**Table 11**

Results in terms of HV obtained by different algorithms applied to CEC2009 test functions.

| Functions |      | Algorithms |          |          |          |
|-----------|------|------------|----------|----------|----------|
|           |      | MOWAA      | MOBSLO   | MOAVOA   | MOGOA    |
| UF1       | Ave  | 5.77E-01   | 4.34E-01 | 7.46E-01 | 3.11E-01 |
|           | Rank | 2          | 3        | 1        | 4        |
|           | Std  | 6.67E-02   | 8.05E-02 | 5.23E-02 | 1.05E-01 |
| UF2       | Ave  | 6.57E-01   | 6.52E-01 | 6.47E-01 | 5.41E-01 |
|           | Rank | 1          | 2        | 3        | 4        |
|           | Std  | 4.63E-02   | 4.86E-02 | 5.30E-02 | 4.51E-02 |
| UF3       | Ave  | 2.00E-01   | 3.29E-01 | 2.89E-01 | 7.27E-02 |
|           | Rank | 3          | 1        | 2        | 4        |
|           | Std  | 2.85E-02   | 4.88E-02 | 2.86E-02 | 7.41E-03 |
| UF4       | Ave  | 3.05E-01   | 2.63E-01 | 2.59E-01 | 2.53E-01 |
|           | Rank | 1          | 2        | 3        | 4        |
|           | Std  | 1.83E-02   | 1.35E-02 | 1.33E-02 | 1.52E-02 |
| UF5       | Ave  | 2.58E-01   | 4.60E-01 | 4.34E-01 | 1.00E-01 |
|           | Rank | 3          | 1        | 2        | 4        |
|           | Std  | 1.43E-01   | 1.68E-01 | 7.62E-02 | 9.42E-03 |
| UF6       | Ave  | 3.14E-01   | 3.70E-01 | 6.35E-01 | 1.23E-01 |
|           | Rank | 3          | 2        | 1        | 4        |
|           | Std  | 2.04E-01   | 1.83E-01 | 1.05E-01 | 1.31E-01 |
| UF7       | Ave  | 4.80E-01   | 4.76E-01 | 6.61E-01 | 3.62E-01 |
|           | Rank | 2          | 3        | 1        | 4        |
|           | Std  | 5.67E-02   | 8.40E-02 | 8.75E-02 | 2.12E-01 |
| UF8       | Ave  | 1.95E-01   | 1.61E-01 | 5.78E-01 | 5.25E-01 |
|           | Rank | 3          | 4        | 1        | 2        |
|           | Std  | 9.16E-03   | 1.54E-01 | 2.12E-01 | 1.03E-01 |
| UF9       | Ave  | 5.53E-01   | 9.02E-01 | 9.10E-01 | 4.74E-01 |
|           | Rank | 3          | 2        | 1        | 4        |
|           | Std  | 1.40E-01   | 4.62E-02 | 4.41E-02 | 1.07E-01 |
| UF10      | Ave  | 5.24E-01   | 1.45E-01 | 5.20E-01 | 2.80E-01 |
|           | Rank | 1          | 4        | 2        | 3        |
|           | Std  | 1.12E-01   | 5.63E-02 | 1.57E-01 | 4.43E-02 |

between  $x$  and  $\Omega$ ;  $|P^*|$  is the number of points in the set  $P^*$ .

- 2) The  $S$  and  $MS$  are used to evaluate the coverage uniformity of the solution produced by the algorithm.  $S$  measures the variance of the interval of neighboring vectors in the known Pareto front as given by Eq. (22). A smaller value of  $S$  is better, as it indicates that the distances between neighboring solutions are more consistent. In the contrast, maximum spread assesses the extent of distribution and dispersion among solutions within the solution set, reflecting the “diversity” of solutions, as expressed in Eq. (23). A larger value of  $MS$  is indeed better, as it indicates that the solutions are more widely distributed, meaning better coverage of the objective space.

$$S = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (\bar{d} - d_i)^2} \quad (22)$$

$$MS = \sqrt{\sum_{i=1}^o \max(d(a_i, b_i))} \quad (23)$$

Where  $\bar{d}$  is the average value of  $d_i$ ;  $n$  is the number of Pareto optimal solutions;  $a_i$  is the maximum value of the  $i_{th}$  target;  $b_i$  is the minimum value of the  $i_{th}$  target,  $d(a_i, b_i)$  is the Euclidean distance between  $a_i$  and  $b_i$ ;  $o$  is the number of targets.

- 3) The last metric is the HV, which measures the size of the target space covered by all solutions to consider the following three aspects simultaneously: precision, diversity, and cardinality. The larger the value of HV, the better the performance.

$$HV(A, r) = Leb\left(\bigcup_{i=1}^{|A|} h(a_i, r)\right) \quad (24)$$

Where  $Leb$  denotes the Lebesgue measure,  $|A|$  the number of

solutions in, and  $h(a_i, r)$  is the volume covered by the  $i_{th}$  solution in the reference point  $r$ , which is formed with the Nadir values of each objective.

### 3.2. Overview of implemented test functions

The proposed algorithm is applied on both two-objective and three-objective benchmark test functions with varying shapes of Pareto optimal fronts to evaluate its performance in multi-objective optimization. Among them, the two-objective benchmark test functions include the UF1-UF7 from the CEC2009 test functions, and ZDT1-ZDT4 from the ZDT test functions. While the three-objective benchmark test functions consist of UF8-UF10 from CEC2009, and DTLZ2, DETLZ6 from the DTLZ benchmark functions. Owing to their characteristics including continuous, concave, and convex fronts, these benchmark functions are regarded as an effective tool to validate the optimization performance for the proposed algorithm. The detailed expressions of these benchmark functions are presented in Appendix A and Appendix B. In addition, the Friedman mean rank test and Wilcoxon signed ranks test are employed to statistically compare the performance of algorithms in a statistical manner.

### 3.3. Comparison of different multi-objective weight average algorithm versions

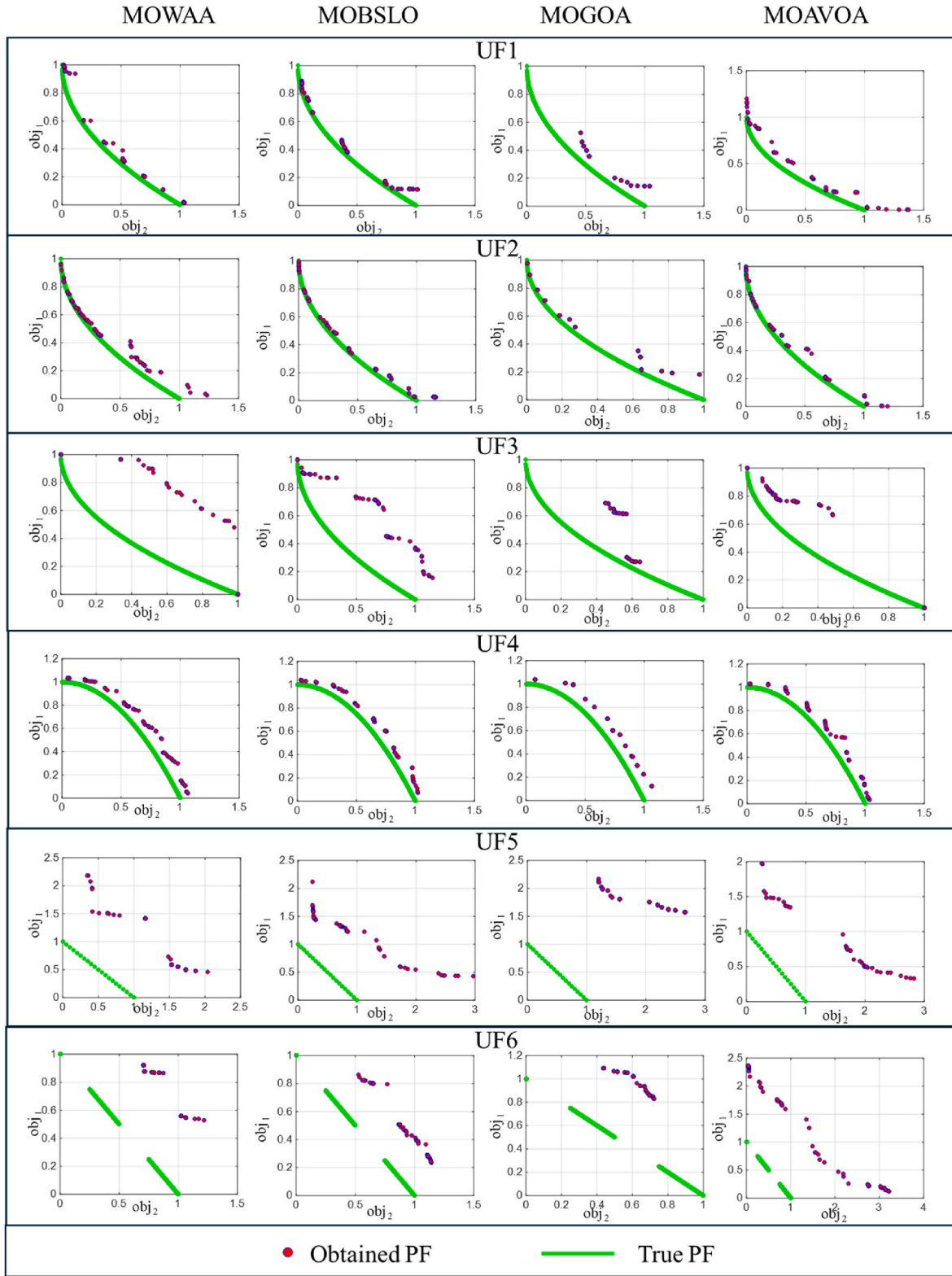
In this section, the impact of each integrated techniques—specifically, the crossover and mutation operations, on the proposed algorithm is studied through the comparison of CEC2009 results. The following configurations are considered.

1. Original MOWAA without any integrated technique, named “MOWAA\_O”.
2. MOWAA\_O with integrated crossover operation, named “MOWAA\_C”.
3. MOWAA\_O with integrated mutation operation, named “MOWAA\_M”.
4. MOWAA\_O with integrated crossover and mutation operations, named “MOWAA”.

Besides, for a fair comparison, the optimization algorithm parameters for the CEC2009 benchmark tests are set as follows: the number of populations  $N = 50$ , the maximum number of iterations  $Iter_{max} = 100$ , and the size of the archive population  $N_{Archive} = 100$ . In addition, to limit the impact of random factors in the algorithms, each configuration is executed independently 30 times, and the mean and standard deviation of the results are reported.

Table 1 and Table 2 present the performance in terms of GD and IGD metrics, respectively for the CEC2009 test functions. The findings clearly reveal that the hybrid integrated MOWAA configuration outperforms the other versions. Specifically, MOWAA obtained the best GD scores for seven CEC2009 benchmark test functions (i.e., UF1-UF3, UF6, UF7 and UF9-UF10) and the best IGD scores for eight of the CEC2009 benchmark test functions (i.e., UF1-UF3 and UF6-UF10). To demonstrate the superiority of MOWAA, Fig. 2 (a) and Fig. 2 (b) illustrate the average IGD and GD values within 30 independent runs on CEC2009 test functions obtained by different versions, respectively. Overall, MOWAA consistently demonstrates the lowest average IGD and GD values. In conclusion, these results highlight the strong convergence and diversity performance of MOWAA on the CEC2009 benchmark test functions.

To evaluate the coverage uniformity of the solutions obtained by the algorithm across ten CEC2009 benchmark test functions, the  $MS$  and  $S$  metrics are reported in Table 3 and Table 4, respectively. The results evidently indicate the superiority of the hybrid integrated version (MOWAA) on these benchmark test functions. MOWAA dominates the other algorithm versions for six out of ten functions based on the  $S$  metric and for seven out of ten functions based on the  $MS$  metric. The



**Fig. 5.** The population distributions of the true Pareto front solutions and the final solutions obtained for CEC2009 benchmark functions using four algorithms including MOALO, MOGWO, MOGOA, and MOWAA. (Legend: Obtained PF (obtained Pareto front solutions); True PF (true Pareto front solutions)).

average MS and S values over 10 independent runs are illustrated in Fig. 2 (c) and Fig. 2 (d). These figures further validate the effectiveness of MOWAA, along with its variants (MOWAA\_M, MOWAA\_C, and MOWAA\_O), in solving complex multi-objective optimization problems.

Finally, the HV metric results for the proposed MOWAA algorithm are presented in Table 5 and compared to those of the other versions. The results indicate that the MOWAA algorithm achieves superior performance, particularly for the test functions UF1-UF4, UF6-UF7, and

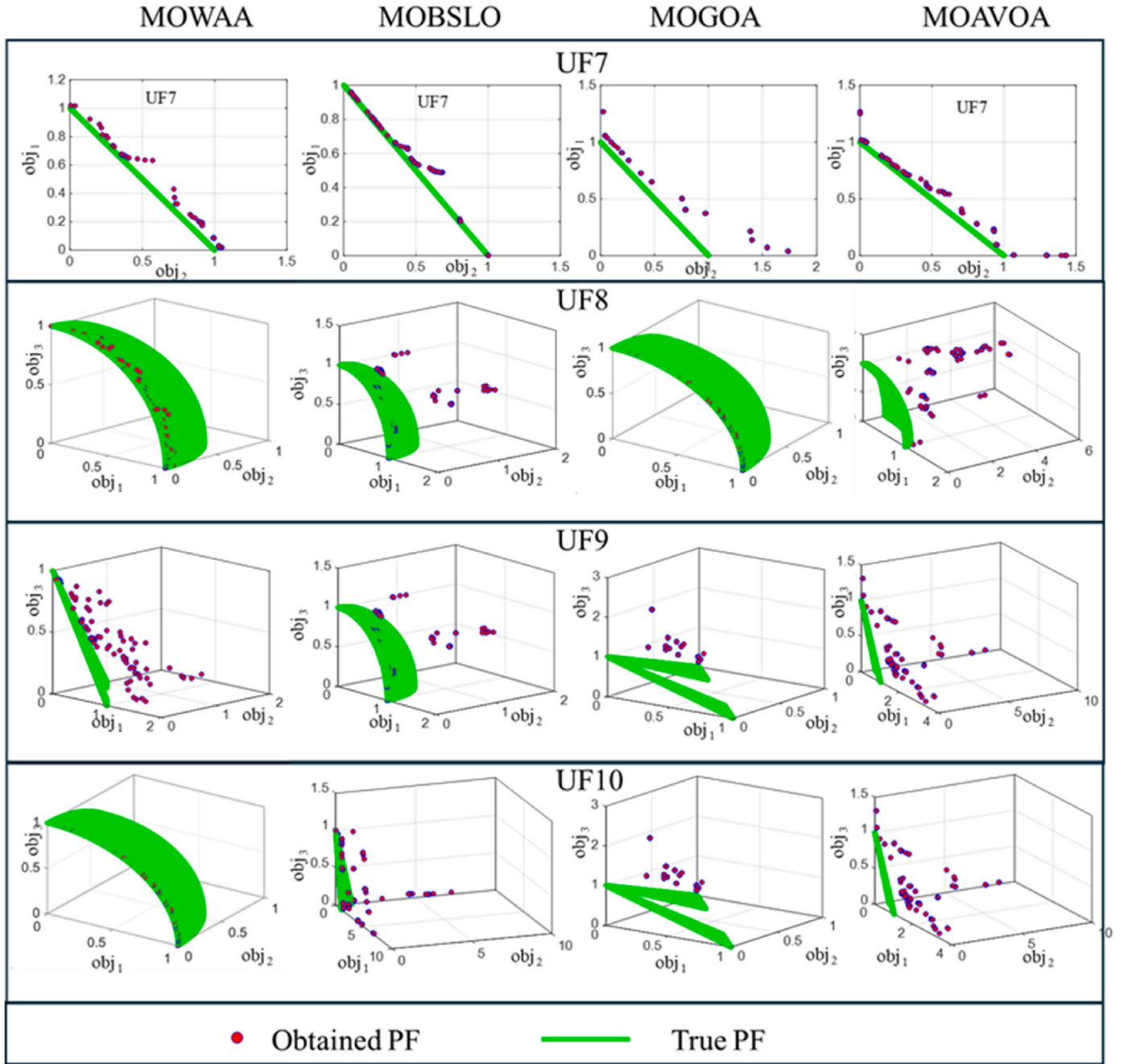


Fig. 5. (continued).

UF9-UF10. The hybrid integration applied to MOWAA\_O obviously improved the results compared to other integration techniques. Overall, MOWAA is identified as the optimal configuration, delivering the best results across the majority of the CEC2009 benchmark test functions. Furthermore, the analysis of Fig. 2(e) reinforces this conclusion, demonstrating that MOWAA exhibits outstanding performance, particularly in terms of the convergence uniformity of solutions.

In general, based on the results presented above, it can be concluded that MOWAA (i.e., the version with integrated crossover and mutation operations) outperforms MOWAA\_O, MOWAA\_C and MOWAA\_M in terms of five different metrics. Therefore, MOWAA will be used to make a broad comparison with some of the most common multi-objective algorithms in section 3.5.

### 3.4. Parameter sensitivity analysis

In this section, the effects of different parameter settings on the performance of the proposed MOWAA algorithm are investigated.

The first parameter examined is the number of iterations, which determines how many times that the mutation operation and crossover operation are executed to construct a front of the non-dominated solutions. These non-dominated solutions, in turn, guide the movements in subsequent iterations to discover additional high-quality solutions. Values of 5, 10, 15, 20, 25, 30, 40, and 50 iterations are tested using a one-way ANOVA with 95 % as a confidence interval across 20 independent runs.

As shown in Fig. 3, there is no consistent relationship between the number of iterations (*nIter*) and the IGD values for UF1 to UF7. To be specific, for UF2 and UF4, a larger number of iterations results in smaller

**Table 12**

$p$  values of the Wilcoxon signed ranks test results used to compare MOWAA with other algorithms for CEC2009 benchmark functions at  $\alpha = 0.05$ .

| Metrics | Comparison      | R <sup>+</sup> | R <sup>-</sup> | p value  | h     |
|---------|-----------------|----------------|----------------|----------|-------|
| IGD     | MOWAA vs MOBSLO | 37             | 18             | 3.75E-01 | FALSE |
|         | MOWAA vs MOAVOA | 24             | 31             | 7.70E-01 | FALSE |
|         | MOWAA vs MOGOA  | 11             | 44             | 1.37E-02 | TRUE  |
| GD      | MOWAA vs MOBSLO | 22             | 33             | 6.25E-01 | FALSE |
|         | MOWAA vs MOAVOA | 5              | 50             | 1.95E-02 | TRUE  |
|         | MOWAA vs MOGOA  | 8              | 47             | 1.95E-02 | TRUE  |
| HV      | MOWAA vs MOBSLO | 27             | 28             | 1.00E+00 | FALSE |
|         | MOWAA vs MOAVOA | 6              | 49             | 2.73E-02 | TRUE  |
|         | MOWAA vs MOGOA  | 45             | 10             | 8.40E-02 | FALSE |
| MS      | MOWAA vs MOBSLO | 29             | 26             | 9.22E-01 | FALSE |
|         | MOWAA vs MOAVOA | 19             | 36             | 4.32E-01 | FALSE |
|         | MOWAA vs MOGOA  | 55             | 0              | 1.95E-03 | TRUE  |
| S       | MOWAA vs MOBSLO | 15             | 40             | 2.32E-01 | FALSE |
|         | MOWAA vs MOAVOA | 7              | 48             | 3.71E-02 | TRUE  |
|         | MOWAA vs MOGOA  | 9              | 46             | 1.95E-02 | TRUE  |

**Table 13**

Results in terms of IGD obtained by different algorithms applied to ZDT and DTLZ test functions.

| Functions |      | Algorithms |          |          |          |
|-----------|------|------------|----------|----------|----------|
|           |      | MOWAA      | MOBSLO   | MOAVOA   | MOGOA    |
| ZDT1      | Ave  | 1.48E-03   | 2.28E-03 | 1.37E-03 | 4.99E-02 |
|           | Rank | 2          | 3        | 1        | 4        |
|           | Std  | 2.80E-04   | 2.74E-04 | 2.53E-04 | 5.24E-03 |
| ZDT2      | Ave  | 5.72E-04   | 2.11E-03 | 1.49E-03 | 5.40E-02 |
|           | Rank | 1          | 3        | 2        | 4        |
|           | Std  | 3.80E-04   | 2.67E-04 | 2.34E-04 | 5.48E-03 |
| ZDT3      | Ave  | 1.24E-02   | 1.25E-02 | 1.25E-02 | 4.43E-02 |
|           | Rank | 1          | 2        | 3        | 4        |
|           | Std  | 3.40E-04   | 1.23E-04 | 1.35E-04 | 1.60E-03 |
| ZDT4      | Ave  | 1.88E-02   | 1.88E-02 | 1.88E-02 | 8.70E+00 |
|           | Rank | 1          | 3        | 2        | 4        |
|           | Std  | 7.04E-05   | 1.05E-04 | 7.08E-05 | 9.69E-02 |
| DTLZ2     | Ave  | 1.02E-03   | 2.35E-03 | 9.32E-04 | 2.95E-03 |
|           | Rank | 2          | 3        | 1        | 4        |
|           | Std  | 5.61E-05   | 5.94E-04 | 6.81E-05 | 7.19E-04 |
| DTLZ4     | Ave  | 1.56E-03   | 2.10E-03 | 1.70E-03 | 1.38E-02 |
|           | Rank | 1          | 3        | 2        | 4        |
|           | Std  | 2.35E-04   | 7.84E-04 | 5.25E-04 | 1.24E-03 |

**Table 14**

Results in terms of GD obtained by different algorithms applied to ZDT and DTLZ test functions.

| Functions |      | Algorithms |          |          |          |
|-----------|------|------------|----------|----------|----------|
|           |      | MOWAA      | MOBSLO   | MOAVOA   | MOGOA    |
| ZDT1      | Ave  | 9.54E-04   | 2.36E-04 | 9.96E-03 | 9.82E-02 |
|           | Rank | 2          | 1        | 3        | 4        |
|           | Std  | 1.83E-04   | 8.39E-05 | 3.18E-02 | 2.73E-02 |
| ZDT2      | Ave  | 3.86E-04   | 9.48E-05 | 4.40E-04 | 1.09E-01 |
|           | Rank | 2          | 1        | 3        | 4        |
|           | Std  | 1.88E-04   | 1.15E-05 | 1.05E-03 | 8.54E-03 |
| ZDT3      | Ave  | 3.48E-04   | 4.61E-04 | 1.08E-02 | 9.26E-02 |
|           | Rank | 1          | 2        | 3        | 4        |
|           | Std  | 2.00E-04   | 1.42E-04 | 1.97E-02 | 1.53E-02 |
| ZDT4      | Ave  | 5.54E-04   | 4.79E-04 | 1.81E-01 | 1.98E+01 |
|           | Rank | 2          | 1        | 3        | 4        |
|           | Std  | 2.97E-04   | 2.34E-04 | 5.20E-01 | 1.49E-01 |
| DTLZ2     | Ave  | 3.17E-03   | 2.30E-01 | 1.38E-02 | 5.86E-04 |
|           | Rank | 2          | 4        | 3        | 1        |
|           | Std  | 1.95E-04   | 8.59E-02 | 1.41E-02 | 7.22E-05 |
| DTLZ4     | Ave  | 3.95E-03   | 4.01E-01 | 7.11E-01 | 1.85E-01 |
|           | Rank | 1          | 3        | 4        | 2        |
|           | Std  | 1.53E-03   | 2.43E-01 | 3.18E-01 | 1.19E-01 |

**Table 15**

Results in terms of MS obtained by different algorithms applied to ZDT and DTLZ test functions.

| Functions |      | Algorithms |          |          |          |
|-----------|------|------------|----------|----------|----------|
|           |      | MOWAA      | MOBSLO   | MOAVOA   | MOGOA    |
| ZDT1      | Ave  | 1.03E+00   | 9.58E-01 | 1.21E+00 | 1.00E+00 |
|           | Rank | 2          | 4        | 1        | 3        |
|           | Std  | 3.16E-02   | 2.99E-02 | 7.09E-01 | 5.44E-01 |
| ZDT2      | Ave  | 1.03E+00   | 9.75E-01 | 1.00E+00 | 9.52E-01 |
|           | Rank | 1          | 3        | 2        | 4        |
|           | Std  | 3.69E-02   | 2.05E-02 | 1.71E-02 | 3.25E-01 |
| ZDT3      | Ave  | 1.01E+00   | 9.88E-01 | 1.19E+00 | 1.52E+00 |
|           | Rank | 3          | 4        | 2        | 1        |
|           | Std  | 1.15E-02   | 1.08E-02 | 4.02E-01 | 2.40E-01 |
| ZDT4      | Ave  | 1.02E+00   | 9.66E-01 | 4.25E+00 | 1.00E+00 |
|           | Rank | 2          | 4        | 1        | 3        |
|           | Std  | 5.72E-03   | 2.82E-02 | 9.57E+00 | 6.72E-01 |
| DTLZ2     | Ave  | 9.59E-01   | 7.93E+00 | 1.39E+00 | 1.05E+00 |
|           | Rank | 4          | 1        | 2        | 3        |
|           | Std  | 1.55E-02   | 1.45E+00 | 4.84E-01 | 1.69E-01 |
| DTLZ4     | Ave  | 9.52E-01   | 9.96E+00 | 1.26E+01 | 5.05E-01 |
|           | Rank | 3          | 2        | 1        | 4        |
|           | Std  | 3.45E-02   | 1.98E+00 | 3.69E+00 | 7.15E-01 |

**Table 16**

Results in terms of S obtained by different algorithms applied to ZDT and DTLZ test functions.

| Functions |      | Algorithms |          |          |          |
|-----------|------|------------|----------|----------|----------|
|           |      | MOWAA      | MOBSLO   | MOAVOA   | MOGOA    |
| ZDT1      | Ave  | 2.12E-02   | 8.31E-03 | 1.25E-02 | 8.57E-04 |
|           | Rank | 4          | 2        | 3        | 1        |
|           | Std  | 3.86E-03   | 3.07E-03 | 3.22E-03 | 8.04E-04 |
| ZDT2      | Ave  | 2.41E-02   | 6.83E-03 | 1.13E-02 | 1.58E-03 |
|           | Rank | 4          | 1        | 2        | 3        |
|           | Std  | 5.71E-03   | 2.34E-03 | 2.57E-03 | 4.77E-05 |
| ZDT3      | Ave  | 2.72E-02   | 1.41E-02 | 1.53E-02 | 3.13E-04 |
|           | Rank | 4          | 2        | 3        | 1        |
|           | Std  | 5.14E-03   | 4.28E-03 | 2.67E-03 | 1.99E-04 |
| ZDT4      | Ave  | 2.04E-02   | 9.51E-03 | 1.15E-02 | 3.02E-01 |
|           | Rank | 3          | 1        | 2        | 4        |
|           | Std  | 3.79E-03   | 4.93E-03 | 3.02E-03 | 2.55E-01 |
| DTLZ2     | Ave  | 7.65E-02   | 3.23E-01 | 9.43E-02 | 7.15E-03 |
|           | Rank | 2          | 4        | 3        | 1        |
|           | Std  | 5.44E-03   | 4.14E-01 | 2.78E-02 | 5.55E-03 |
| DTLZ4     | Ave  | 7.78E-02   | 3.61E-01 | 4.23E-01 | 8.29E-02 |
|           | Rank | 1          | 3        | 4        | 2        |
|           | Std  | 1.68E-02   | 3.21E-01 | 3.86E-01 | 7.21E-02 |

IGD values, with the best performance achieved at 50 iterations. For the other benchmark functions, the IGD values firstly increase from 5 to 25 iterations, and then decrease as the number of iterations approaches 50 iterations. Thus, for these problems, the best results are generally observed either at 5 or 50 iterations. Based on this analysis, larger iteration results in better optimization performances, therefore  $N_{iteration} = 3000$  is selected for subsequent experiments, as it yields better performance across most of the evaluated functions.

The second parameter investigated is  $N$ , which determines the number of selected individuals from the archive used for local search to discover additional non-dominated solutions. The investigation is conducted for  $N$  values of 10, 20, 30, 40, 50, 60, 70, 80, 90, and 100 with the results depicted in Fig. 4. It can be observed that for most test functions, UF1-UF4 and UF6-UF7, a larger population size generally leads to lower IGD values. This indicates that the proposed algorithm performs better when  $N = 100$  for most of the problems. Accordingly, the population size is set to 100 in the subsequent experiments.

After completing the parameter analysis, the superiority of the proposed algorithm will be evaluated in the next section through a comparison with several state-of-the-art multi-objective optimization algorithms.

**Table 17**

Results in terms of HV obtained by different algorithms applied to ZDT and DTLZ test functions.

| Functions |      | Algorithms |          |          |          |
|-----------|------|------------|----------|----------|----------|
|           |      | MOWAA      | MOBSLO   | MOAVOA   | MOGOA    |
| ZDT1      | Ave  | 6.16E-01   | 5.99E-01 | 6.53E-01 | 2.06E-01 |
|           | Rank | 2          | 3        | 1        | 4        |
|           | Std  | 2.13E-02   | 2.11E-02 | 8.87E-02 | 2.26E-02 |
| ZDT2      | Ave  | 2.93E-01   | 2.78E-01 | 3.14E-01 | 1.34E-01 |
|           | Rank | 2          | 3        | 1        | 4        |
|           | Std  | 2.40E-02   | 1.59E-02 | 2.13E-02 | 6.10E-02 |
| ZDT3      | Ave  | 7.35E-01   | 7.34E-01 | 7.82E-01 | 2.44E-01 |
|           | Rank | 2          | 3        | 1        | 4        |
|           | Std  | 1.84E-02   | 1.80E-02 | 7.36E-02 | 1.72E-02 |
| ZDT4      | Ave  | 6.39E-01   | 6.08E-01 | 6.90E-01 | 2.50E-02 |
|           | Rank | 2          | 3        | 1        | 4        |
|           | Std  | 1.46E-02   | 2.00E-02 | 1.29E-01 | 6.98E-03 |
| DTLZ2     | Ave  | 4.54E-01   | 8.79E-01 | 5.93E-01 | 1.24E-01 |
|           | Rank | 3          | 1        | 2        | 4        |
|           | Std  | 3.59E-02   | 6.16E-02 | 1.39E-01 | 4.01E-02 |
| DTLZ4     | Ave  | 4.62E-01   | 9.86E-01 | 9.93E-01 | 3.62E-01 |
|           | Rank | 3          | 2        | 1        | 4        |
|           | Std  | 6.62E-02   | 1.34E-02 | 1.09E-02 | 2.49E-01 |

### 3.5. Comparison of the performance of multi-objective weight average algorithm with other multi-objective optimization algorithms

In this section, the performance of MOWAA is compared with MOGOA, MOBSLO and MOAVOA on ten CEC2009 benchmark test functions (UF1–UF10), four ZDT benchmark functions (ZDT1–ZDT4), two DTLZ benchmark functions (DTLZ2, DTLZ4) and four engineering problems. The algorithm's parameters are detailed in Table 6. Besides, for a fair comparison, the parameters of optimization used in these test functions are identical, i.e., the population size  $N = 100$ , the archive size  $N_{Archive} = 100$ , and the maximum number of iterations  $Iter_{max} = 3000$ . In addition, 30 independent runs of each algorithm are performed to limit the impact of random factors. The mean value and the standard deviation of the IGD, S, MS and HV metrics are calculated to quantify the performance of MOWAA.

#### 3.5.1. Comparison of the algorithms for test functions of CEC 2009

Tables 7 and 8 present the results of the CEC2009 benchmark test functions in terms of GD and IGD performance metrics, respectively. The results clearly indicate that the proposed MOWAA algorithm outperforms the competitive algorithms (MOBSLO, MOGOA, and MOAVOA) in terms of both IGD and GD metrics. Specifically, for GD, MOWAA achieved the best performance for five out of ten test functions (UF2, UF7, UF8, UF9, and UF10), and the second-best performance for three functions (UF1, UF5, and UF6). Although MOBSLO yielded the top GD performance for UF1, UF5, and UF6, the differences between its results and those of MOWAA are negligible. MOAVOA achieved the best GD results for UF3 and UF4, with improvements of approximately 13.0 % and 6.6 %, respectively. Regarding IGD performance, MOWAA ranked first for two out of ten functions (UF1 and UF7), and second for five functions (UF5, UF6, UF8, UF9, and UF10). For the remaining functions, MOAVOA achieved the best results for UF3 and UF4, while MOBSLO obtained the best result for UF2. However, the differences between the best-performing results and those achieved by MOWAA are minimal.

Overall, these findings suggest that the Pareto front solutions generated by MOWAA are closer to the true Pareto front compared to those obtained by the other algorithms. This indicates that MOWAA demonstrates superior convergence capabilities, effectively balancing exploration and exploitation within the search space. These advantages are primarily attributed to the integration of the single-objective Weighted Aggregation Algorithm (WAA) alongside mutation and crossover operators, which collectively enhance the algorithm's global search ability and help prevent premature convergence. This hybrid

update mechanism enables MOWAA to explore the Pareto front more efficiently while maintaining a high-quality approximation to the true front, as evidenced by its consistently strong performance across various benchmark functions.

Tables 9 and 10 present the results of the CEC2009 benchmark test functions based on the MS and S performance metrics. The results clearly demonstrate that the proposed MOWAA outperforms the other algorithms overall. In terms of MS, MOWAA ranks first for two out of ten test functions (UF5 and UF10) and second-best for three functions (UF1, UF3, and UF8). MOAVOA, meanwhile, achieves the best performance in the remaining functions. With respect to the S metric, MOWAA ranks first in six out of ten test functions; all except UF2, UF3, UF6, and UF7. Although MOBSLO shows better performance in UF2, UF6, and UF7, the differences compared to MOWAA are negligible. In addition, MOAVOA ranks first for UF3 in terms of S, but the improvement over MOWAA is minimal, with a relative difference of approximately 18 %. Overall, these results indicate that MOWAA demonstrates superior diversity and distribution uniformity, as evidenced by its consistently strong performance in both MS and S metrics across the majority of test functions.

Table 11 presents the results for the CEC2009 benchmark test functions based on the HV performance metric. It is evident that MOWAA outperforms the other methods overall. Specifically, MOWAA ranks first for three out of ten test functions (UF2, UF4, and UF10), and second-best for two functions (UF1 and UF7). MOAVOA achieves the best HV results for UF1, UF6, UF7, UF8, and UF9, while MOBSLO performs best for UF3 and UF5. However, the differences between the optimal results obtained by these algorithms and those achieved by MOWAA are minimal, indicating that their performances are at the same level in those cases.

In conclusion, the consistently high HV values obtained by MOWAA across the benchmark functions demonstrate its strong capability in maintaining a well-distributed and convergent set of solutions, further confirming its effectiveness in solving complex multi-objective optimization problems.

Based on the visual results illustrated in Fig. 5 for the CEC2009 benchmark problems, it is evident that the proposed MOWAA algorithm demonstrates superior performance compared to the other three algorithms—MOBSLO, MOGOA, and MOAVOA—particularly in terms of solution distribution, spread, and convergence toward the true Pareto front (PF). The comparison encompasses both two-objective and three-objective optimization scenarios, providing a comprehensive evaluation.

For the two-objective problems (UF1–UF6), MOWAA consistently shows a more uniform and dense distribution of obtained Pareto front solutions along the curve of the true Pareto front. Specifically, for functions UF1, UF2, UF4, and UF7, MOWAA achieves an almost perfect alignment with the true PF, indicating not only high convergence accuracy but also superior diversity across the front. In contrast, for functions UF3, UF5, and UF6, while a noticeable gap between the obtained and true Pareto fronts remains, the deviation is significantly smaller compared to the gaps observed in the results of the other algorithms. This suggests that even in more complex scenarios, MOWAA maintains closer proximity to the optimal front, highlighting its robustness. For the three-objective problems (UF7–UF10), MOWAA again excels by producing solution sets that are both well-distributed and well-converged on the true Pareto surface. The 3D plots clearly show that the Pareto front obtained by MOWAA closely adheres to the shape and boundaries of the known true PF, indicating its effectiveness in handling higher-dimensional objective spaces. The quality of distribution is visually evident in the uniform coverage of the front without significant clustering or gaps, which is not consistently observed in the outputs of MOBSLO, MOGOA, or MOAVOA. In summary, the graphical comparison reveals that MOWAA not only converges more closely to the true PF but also maintains a high level of solution diversity and spread, regardless of the problem dimensionality. These qualities affirm the robustness, adaptability, and effectiveness of the MOWAA algorithm in

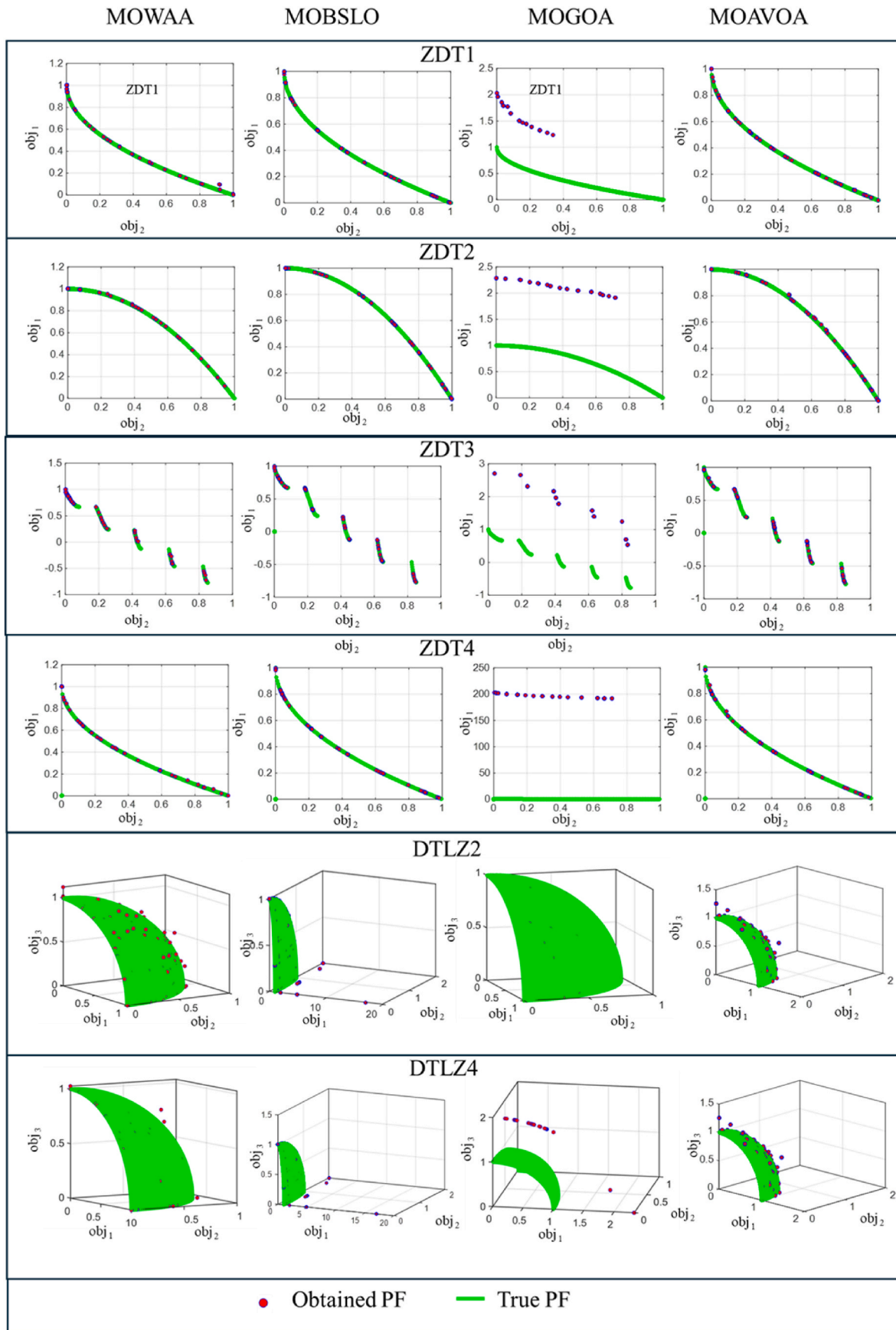


Fig. 6. The population distributions of the true Pareto front solutions and the final solutions obtained for DTLZ and ZDT benchmark functions using four algorithms including MOGOA, MOWAA, MOGOA and MOALO. (Legend: Obtained PF (obtained Pareto front solutions); True PF (true Pareto front solutions)).

**Table 18**

$p$  values of the Wilcoxon signed ranks test used to compare MOWAA with other algorithms for ZDT and DTLZ test function at  $\alpha = 0.05$ .

| Metrics | Comparison      | R <sup>+</sup> | R <sup>-</sup> | p value  | h     |
|---------|-----------------|----------------|----------------|----------|-------|
| IGD     | MOWAA vs MOBSLO | 0              | 21             | 3.13E-02 | TRUE  |
|         | MOWAA vs MOAVOA | 5              | 16             | 3.13E-01 | FALSE |
|         | MOWAA vs MOGOA  | 0              | 21             | 3.13E-02 | TRUE  |
| GD      | MOWAA vs MOBSLO | 8              | 13             | 6.88E-01 | FALSE |
|         | MOWAA vs MOAVOA | 0              | 21             | 3.13E-02 | TRUE  |
|         | MOWAA vs MOGOA  | 1              | 20             | 6.25E-02 | FALSE |
| HV      | MOWAA vs MOBSLO | 10             | 11             | 1.00E+00 | FALSE |
|         | MOWAA vs MOAVOA | 0              | 21             | 3.13E-02 | TRUE  |
|         | MOWAA vs MOGOA  | 21             | 0              | 3.13E-02 | TRUE  |
| MS      | MOWAA vs MOBSLO | 10             | 11             | 1.00E+00 | FALSE |
|         | MOWAA vs MOAVOA | 1              | 16             | 6.25E-02 | FALSE |
|         | MOWAA vs MOGOA  | 11             | 10             | 1.00E+00 | FALSE |
| S       | MOWAA vs MOBSLO | 10             | 11             | 1.00E+00 | FALSE |
|         | MOWAA vs MOAVOA | 10             | 11             | 1.00E+00 | FALSE |
|         | MOWAA vs MOGOA  | 14             | 7              | 5.63E-01 | FALSE |

addressing complex multi-objective optimization problems across the CEC2009 benchmark suite.

The results presented in previous tables and figures indicate that MOWAA has achieved remarkable success in generating near-optimal Pareto solutions, surpassing other algorithms in both quality and quantity. Furthermore, the Wilcoxon signed-rank test is applied to compare the performance of two algorithms for the same benchmark function, aiming to determine whether there is a statistically significant difference in performance based on their average values. If the test result  $h$  is TRUE, it indicates a significant difference between the paired algorithms, whereas  $h$  being FALSE suggests no significant difference in their performance. According to the results in Table 12, MOWAA significantly outperforms the other algorithms, underscoring its superiority in terms of both performance and effectiveness. Overall, MOWAA presents itself as a promising and effective multi-objective optimization algorithm for producing near-optimal Pareto solutions of high quality and quantity.

### 3.5.2. Discussion of the DTLZ and ZDT functions

The advanced multi-modal benchmark functions with fixed dimensions, including ZDT (i.e., ZDT1-ZDT6) and DTLZ (DTLZ2 and DTLZ4), are used in this subsection to further evaluate the performance of all algorithms including MOWAA, MOAVOA, MOGOA and MOBSLO. The descriptions of these multi-objective benchmark test functions are presented in Appendix B.1 and Appendix B.2.

The values of the performance metrics GD and IGD for ZDT and DTLZ benchmark test functions are summarized in Table 13 and Table 14. The proposed algorithm MOWAA exhibits competitive and often superior

performance compared to other algorithms. Notably, MOWAA achieves the best average IGD values across all test functions, indicating strong convergence toward and uniform distribution along the true Pareto front. In terms of GD, MOWAA secures the best performance in two out of six cases (ZDT3, and DTLZ4) further supporting its strong convergence capabilities. Additionally, MOWAA ranks second-best in terms of GD for ZDT2, ZDT4, and in terms of both GD and IGD for ZDT1 and DTLZ2, which reinforces its robustness and consistency even in challenging test scenarios. The low standard deviation values across both metrics further validate the algorithm's reliability and stability over multiple runs, highlighting its ability to consistently deliver high-quality solutions.

To evaluate the uniformity of solution coverage achieved by the algorithms, the performance metrics S and MS for the ZDT and DTLZ benchmark test functions are presented in Tables 15 and 16. The results demonstrate that the proposed MOWAA algorithm delivers excellent MS performance, outperforming MOBSLO and MOGOA, though slightly lagging behind MOAVOA. Specifically, MOWAA achieves the best MS value for one of the six functions (ZDT2) and the second-best MS values for two cases (ZDT1 and ZDT4). For the remaining functions, the results obtained by MOWAA are effectively indistinguishable from the optimal values, highlighting the algorithm's robustness in maintaining a well-distributed solution set. Regarding the S metric, which evaluates the evenness of spacing among solutions, MOWAA achieves first-best and second-best performance for two key functions, DTLZ2 and DTLZ4, respectively. Other algorithms show mixed strengths: MOGOA ranks first for ZDT1 and ZDT3, while MOBSLO secures the top position for ZDT2 and ZDT4.

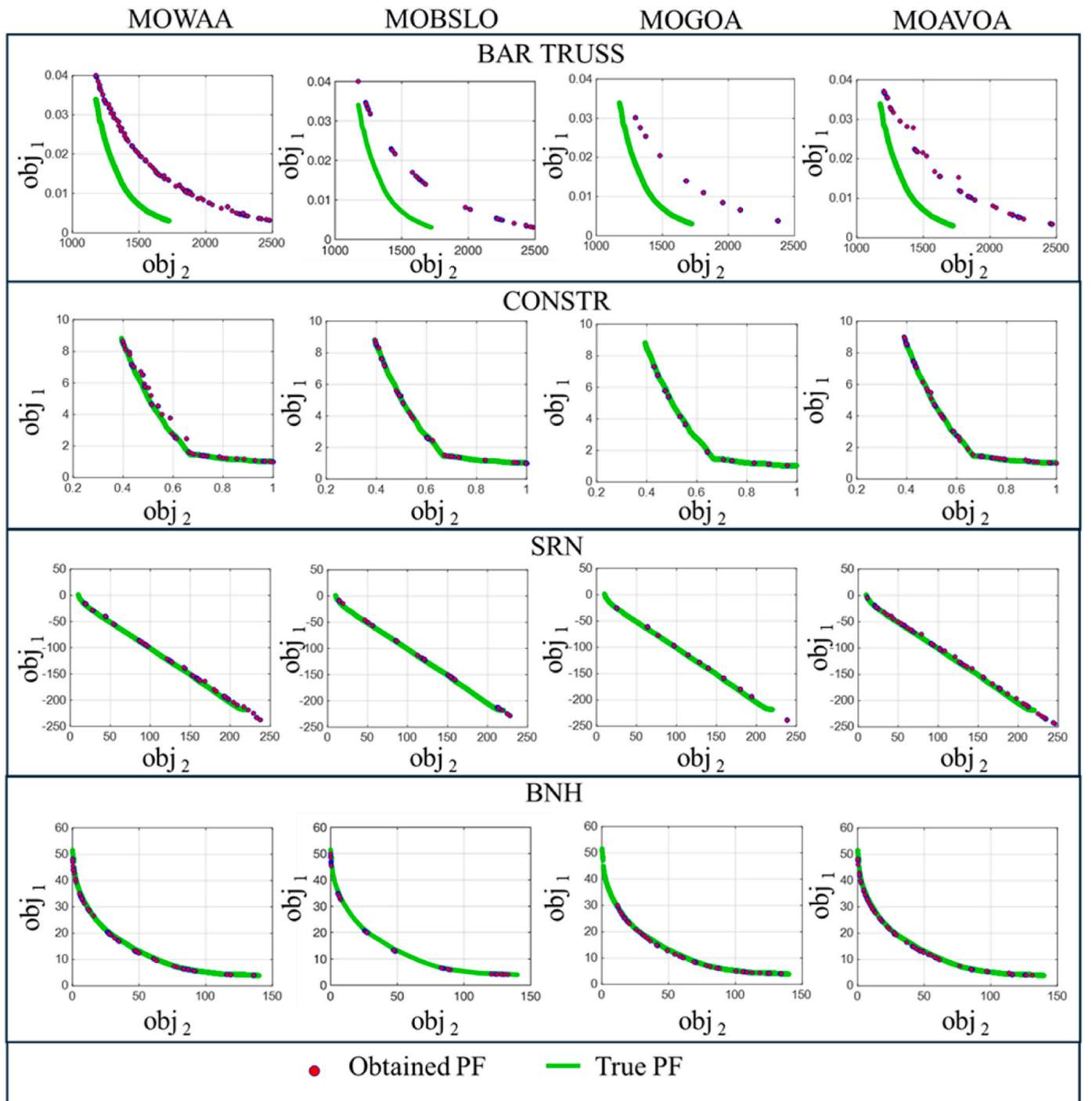
The results for the performance metric HV, as presented in Table 17, indicate that MOAVOA achieves the best performance across five out of six benchmark test functions (ZDT1 through ZDT4 and DTLZ4) making it the top performer in terms of this metric. MOWAA follows closely as the second-best algorithm overall. In particular, MOWAA demonstrates strong HV performance in the ZDT test suite, with relative errors compared to the optimal HV values remaining below 7.35 % for ZDT1, ZDT2, ZDT3, and ZDT4. Its performance in the DTLZ benchmarks is slightly weaker. The relative error for DTLZ2 is 23.4 %, and 53.4 % for DTLZ4, suggesting room for improvement in handling certain high-dimensional objective spaces. Nevertheless, the standard deviation values remain low, indicating that MOWAA consistently delivers stable and repeatable results across multiple runs.

Fig. 6 illustrates the best Pareto fronts obtained by the MOWAA algorithm in comparison with three competitive algorithms—MOAVOA, MOBSLO, and MOGOA—across a suite of ZDT and DTLZ benchmark test functions. From the visualizations, it is evident that MOWAA consistently achieves near-complete convergence to the true Pareto-optimal

**Table 19**

Statistical analysis of the algorithms in solving four engineering problems.

| Engineering Problems | Algorithms | GD       |          | S        |          | IGD      |          | MS       |          | HV       |          |
|----------------------|------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
|                      |            | Ave      | SD       | Ave      | SD       | Ave      | SD       | Ave      | SD       | Ave      | SD       |
| P1: BNH              | MOGOA      | 7.81E-04 | 1.12E-04 | 1.53E+00 | 3.10E-01 | 1.80E-03 | 5.74E-04 | 3.10E-01 | 4.70E-03 | 6.77E-01 | 4.98E-02 |
|                      | MOWAA      | 5.10E-04 | 6.45E-05 | 1.02E+00 | 1.89E-01 | 9.90E-04 | 1.03E-04 | 1.04E+00 | 8.50E-03 | 7.50E-01 | 1.98E-02 |
|                      | MOAVOA     | 5.85E-04 | 5.68E-05 | 1.18E+00 | 2.28E-01 | 7.93E-04 | 2.14E-04 | 9.35E-01 | 2.08E-02 | 7.20E-01 | 1.46E-02 |
|                      | MOBSLO     | 6.39E-04 | 8.67E-05 | 4.77E-01 | 1.88E-01 | 2.00E-03 | 1.42E-04 | 9.56E-01 | 1.04E-02 | 6.92E-01 | 1.16E-02 |
| P2: CONSTR           | MOGOA      | 4.84E-04 | 3.47E-05 | 2.85E-02 | 3.31E-02 | 1.66E-03 | 4.13E-04 | 2.29E-02 | 7.38E-03 | 3.96E-01 | 1.56E-02 |
|                      | MOWAA      | 1.24E-03 | 3.72E-04 | 5.20E-02 | 1.91E-02 | 8.59E-04 | 3.07E-05 | 1.01E+00 | 1.59E-02 | 4.08E-01 | 2.07E-02 |
|                      | MOAVOA     | 7.98E-04 | 2.68E-04 | 6.86E-02 | 1.14E-02 | 8.52E-04 | 1.41E-04 | 9.99E-01 | 2.05E-02 | 4.03E-01 | 1.05E-02 |
|                      | MOBSLO     | 6.34E-02 | 1.03E-02 | 4.12E+00 | 2.73E+00 | 1.02E-02 | 3.13E-04 | 1.90E+00 | 6.16E-02 | 3.19E-01 | 1.61E-02 |
| P3: SRN              | MOGOA      | 5.43E-03 | 3.01E-03 | 1.61E-02 | 7.59E-03 | 1.81E-03 | 2.50E-04 | 6.10E-01 | 2.83E-02 | N/A      | N/A      |
|                      | MOWAA      | 3.25E-03 | 3.76E-04 | 2.45E+00 | 4.87E-01 | 1.12E-03 | 1.12E-04 | 9.77E-01 | 7.77E-03 | N/A      | N/A      |
|                      | MOAVOA     | 4.57E-03 | 1.87E-03 | 3.89E+00 | 5.70E-01 | 8.12E-04 | 1.14E-04 | 1.11E+00 | 3.69E-02 | 4.72E-01 | 4.72E-01 |
|                      | MOBSLO     | 5.17E-04 | 1.13E-04 | 4.59E-02 | 2.09E-02 | 1.54E-03 | 1.77E-04 | 1.00E+00 | 1.45E-02 | 3.95E-01 | 1.34E-02 |
| P4: BAR4TRUSS        | MOGOA      | 1.12E-01 | 7.33E-02 | 3.18E+01 | 4.50E+01 | 1.14E-02 | 1.48E-03 | 9.27E-01 | 1.49E-01 | 2.46E-01 | 3.33E-02 |
|                      | MOWAA      | 6.10E-02 | 6.86E-03 | 1.06E+01 | 1.47E+00 | 9.82E-03 | 5.76E-05 | 6.54E-01 | 8.05E-04 | 3.40E-01 | 1.37E-02 |
|                      | MOAVOA     | 6.02E-02 | 6.07E-03 | 1.07E+01 | 2.25E+00 | 9.96E-03 | 6.87E-05 | 1.90E+00 | 4.31E-02 | 3.26E-01 | 1.96E-02 |
|                      | MOBSLO     | 6.34E-02 | 1.03E-02 | 4.12E+00 | 2.73E+00 | 1.02E-02 | 3.13E-04 | 1.90E+00 | 6.16E-02 | 3.26E-01 | 2.01E-02 |



**Fig. 7.** The population distributions of the true Pareto front solutions and the final solutions obtained for engineering problems using four algorithms including MOGOA, MOWAA, MOAVOA and MOBSLO. (Legend: Obtained PF (obtained Pareto front solutions); True PF (true Pareto front solutions)).

fronts for all test functions. Its solutions not only align closely with the true fronts but also exhibit an exceptionally uniform distribution, underscoring its strong performance in both convergence and diversity. In contrast, MOAVOA exhibits strong convergence performance; however, in several test functions (such as ZDT3 and DTLZ1), a certain degree of solution clustering is observed, indicating room for improvement in terms of diversity. The MOGOA algorithm, on the other hand, demonstrates the poorest convergence behavior overall. Its obtained Pareto fronts fail to adequately approach the true fronts across multiple test functions, making it the weakest performer in this comparison. Meanwhile, the solution set obtained by MOBSLO closely matches the true

Pareto front, demonstrating excellent convergence and distribution across most functions—particularly on ZDT1, ZDT2, DTLZ2, and DTLZ4. Nevertheless, for some functions (e.g., ZDT3 and DTLZ1), sparse regions still exist, suggesting there is room for optimization in handling extreme diversity. In summary, the results in Fig. 6 reaffirm the superior performance of MOWAA. It not only approximates the true Pareto fronts more accurately but also ensures a well-spread and evenly distributed set of solutions, making it a highly effective and reliable algorithm for solving complex multi-objective optimization problems across both two- and three-objective scenarios.

Additionally, the Wilcoxon signed-rank test is employed to perform

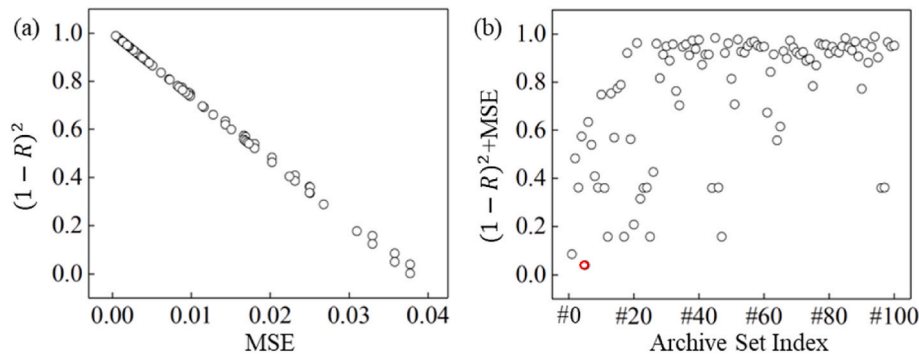


Fig. 8. (a) Performance of Archive set; (b) performance of  $(1 - R)^2 + \text{MSE}$  for the Archive set.

pairwise comparisons of the performance of algorithms applied to the same benchmark problems, aiming to assess whether the differences in their average performance values are statistically significant. A test result of  $h = \text{TRUE}$  indicates a significant difference between the two algorithms, while  $h = \text{FALSE}$  suggests no statistically significant difference. According to the results presented in Table 18, no significant differences are observed between MOWAA and the other algorithms for the MS and S metrics, indicating that all algorithms perform comparably in terms of solution distribution and uniformity. Furthermore, the test reveals the significant differences between MOWAA and MOGOA with respect to the IGD and HV metrics, suggesting different levels of convergence and coverage between these two algorithms. However, significant differences are observed between MOWAA and MOAVOA for the GD and HV metrics, as well as between MOWAA and MOBSLO in terms of IGD, indicating that MOWAA outperforms or behaves distinctly from these algorithms in those aspects. These statistical results reinforce the conclusion that MOWAA delivers competitive and often superior performance, while maintaining consistency across different evaluation metrics and benchmark problems.

Based on the results discussed above, the proposed MOWAA demonstrates remarkable success in generating near-optimal Pareto solutions, surpassing other algorithms in terms of both quality and quantity. However, the statistical analysis indicates that the performance difference between MOWAA and the competitive algorithms is not significant. Therefore, MOWAA can be considered an appealing alternative for solving unconstrained multi-objective problems, offering competitive performance with notable efficiency.

### 3.5.3. Comparison of the algorithms by their application on engineering problems

The above-mentioned benchmark test functions were used to evaluate the performance of MOWAA in solving multi-objective optimization problems without constraint. In this section, several multi-objective engineering problems with constraints are applied for further validation. The detailed descriptions of these engineering problems (BNH, CONSTR, SRN, and BAR TRUSS) are in Appendix C. The algorithm parameters and other parameters including population size, the maximum number of iterations and archive size are shown in Table 6.

The results of the comparison between MOWAA and the algorithms MOGOA, MOAVOA, and MOBSLO in solving engineering problems are presented in Table 19. The performance metrics are defined such that lower values of GD, IGD, and S are preferred, while higher MS and HV values are desirable. Analyzing the average and standard deviation values of these metrics, it is demonstrated that the proposed algorithm MOWAA achieves optimal outcomes for engineering problems BNH, SRN and BAR TRUSS. As for the CONSTR engineering problem, MOAVOA performs best, followed by MOWAA. However, it can be seen that there is no big difference between MOAVOA and MOWAA. Thus, it can be concluded that the proposed MOWAA achieves good performance in solving these real-world engineering problems.

Fig. 7 illustrates the best Pareto-optimal fronts obtained by four algorithms—MOBSLO, MOGOA, MOAVOA, and the proposed MOWAA—when applied to a set of real-world industrial engineering optimization problems. From the visual comparison, it is evident that the MOWAA algorithm consistently yields solutions that are very close to the true Pareto-optimal fronts across all problem instances. This demonstrates its strong convergence capabilities. Moreover, the Pareto fronts produced by MOWAA show excellent alignment with the true optimal boundary and exhibit a smooth and evenly distributed spread of solutions. In contrast, MOGOA and MOAVOA display noticeable deviations, particularly in maintaining uniform coverage across the objective space. MOBSLO, while competitive in some scenarios, often lacks the fine-grained diversity observed in MOWAA's solutions. These observations underscore the robustness and adaptability of MOWAA in handling complex multi-objective optimization problems with real-world constraints. Its superior convergence, along with stable distribution characteristics, make it a promising alternative for industrial applications where both optimality and diversity are critical.

## 4. Application for waviness prediction in wire arc additive manufacturing steel component

Wire arc additive manufacturing (WAAM) has emerged as a highly effective technology for producing large-scale components. However, the process faces significant challenges due to high heat accumulation and complex thermal evolution during deposition, which often lead to severe distortions and dimensional inaccuracies. To address these issues, researchers have employed both simulation and experimental methods to establish mathematical relationships between process parameters and component dimensions. Despite these efforts, the complex interactions among WAAM process parameters, coupled with the high costs of experiments and extensive computational requirements for simulations, have hindered the creation of sufficiently large datasets to fully capture the relationship between process parameters and the resulting component properties.

Recent advancements in computer science have paved the way for surrogate models (SM) based on machine learning techniques, offering faster and more efficient ways to predict and optimize WAAM component geometry. For example, Cheng et al. (Cheng and Waele, 2024b) proposed a method combining artificial neural networks (ANN) with the WAA algorithm to predict and optimize surface waviness in steel components produced by WAAM. Specifically, the Weighted Average Algorithm (WAA) is employed to optimize the initial weights and thresholds of the Artificial Neural Network (ANN), with the Root Mean Square Error (RMSE) serving as the target metric for minimization. This optimization process enhances the ANN's predictive performance by systematically adjusting the network parameters during the initialization phase, ensuring a more effective starting point for subsequent training. By focusing on RMSE reduction, the WAA-ANN approach improves the model's ability to accurately capture complex nonlinear relationships in

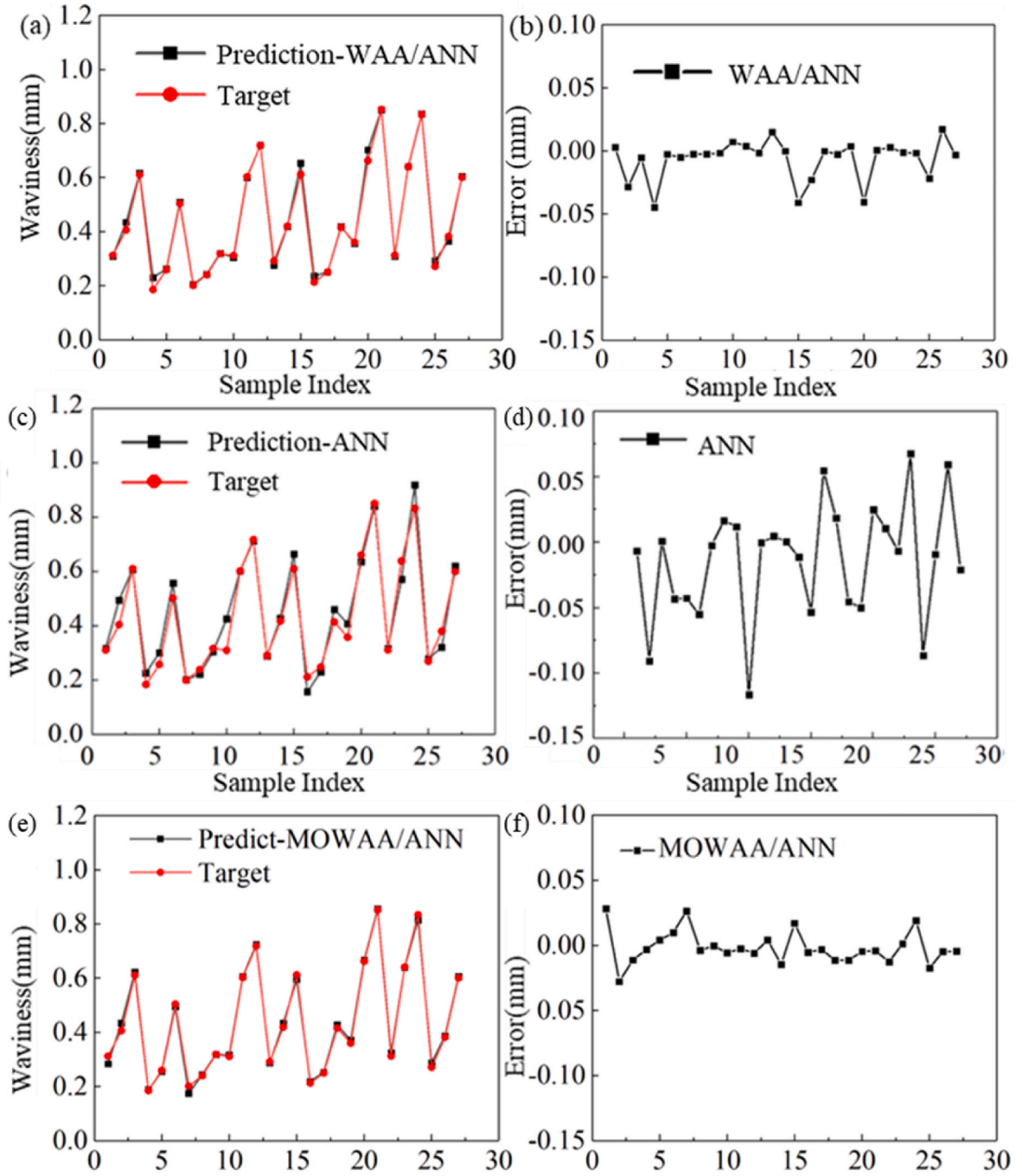


Fig. 9. (a)–(b) Performance of WAA/ANN; (c)–(d) performance of ANN; (e)–(f) performance of MOWAA/ANN.

applications of surface waviness prediction in WAAM.

Despite these advances, most studies have focused on optimizing the hyperparameters of ANN models using single-objective algorithms, such as minimizing *RMSE* or *MSE*. However, this approach often neglects other critical metrics, such as the coefficient of determination (*R*), which measures how well the model explains the variance in the data. Solely optimizing for *RMSE* can result in models with low prediction error but poor explanatory power, as reflected by low *R* values. Therefore, the weighted-sum method (e.g.,  $w_1 \cdot RMSE + (1 - w_1) \cdot (1 - R)^2$ ) are employed as a straightforward way to combine metrics *R* and *RMSE*, however it has inherent limitations: 1) Subjectivity in Weight Selection: The choice of  $w_1$  is arbitrary and may require extensive tuning for different datasets or tasks. 2) Hidden Trade-offs: Weighted sums collapse

multi-dimensional goals into a single value, obscuring potential conflicts between *RMSE* and *R*. Therefore, to overcome this limitation, multi-objective optimization algorithm is proposed as a solution. (multi-objective optimization algorithm) enables simultaneous optimization of *RMSE* and  $(1 - R)^2$  by generating a Pareto front that represents the trade-offs between these conflicting objectives. This multi-objective approach provides greater flexibility to select optimal solutions based on application-specific requirements, improving the balance between prediction accuracy and variance explanation. Compared with the weighted-sum method, (multi-objective optimization algorithm) displays better performance, especially in three aspects.

1. Avoids Subjective Weight Tuning

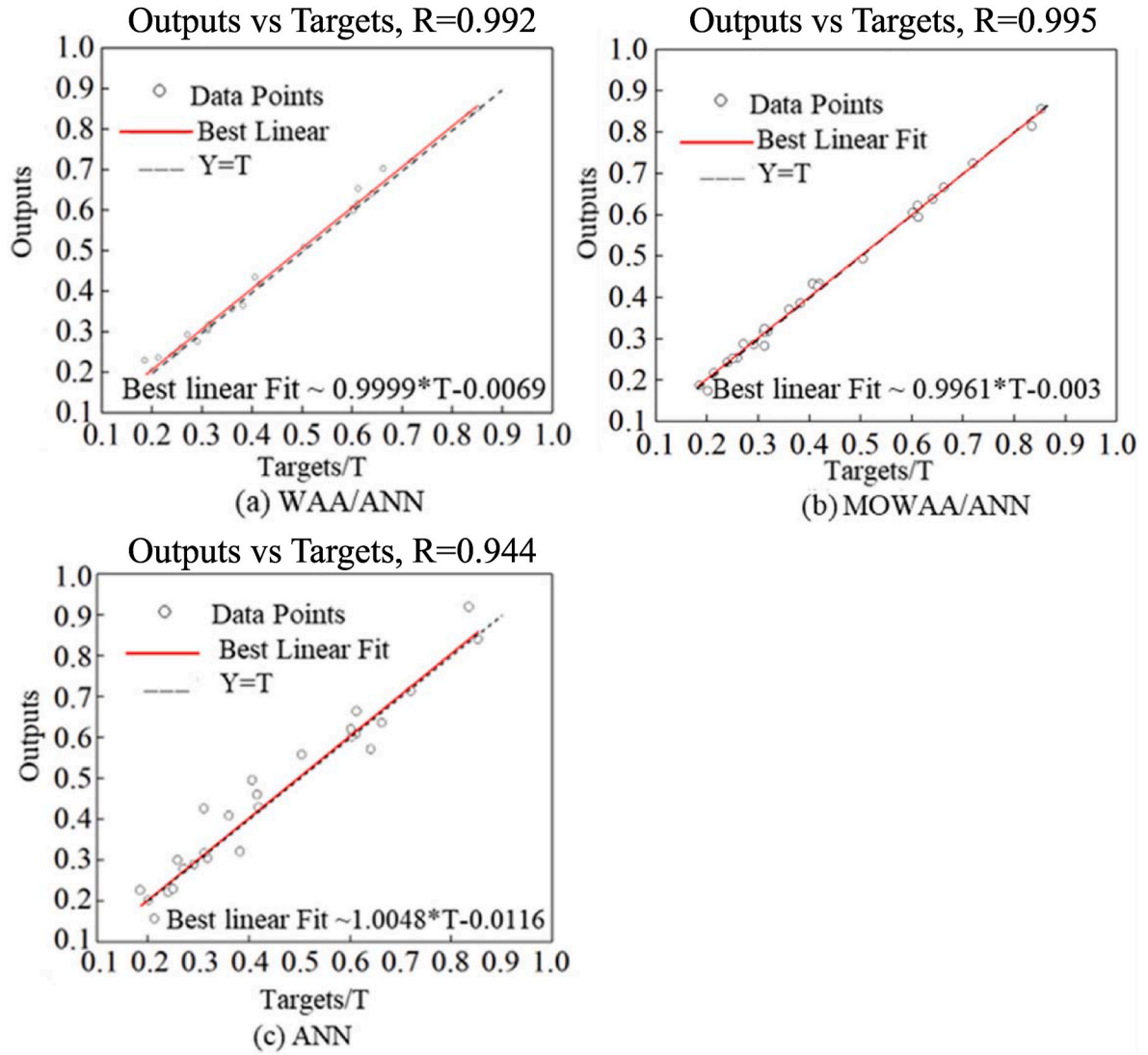


Fig. 10. Regression plots for different prediction models: (a) WAA/ANN; (b) MOWAA/ANN; (c) ANN.

Table 20

Summary of the performance metrics for different waviness prediction models.

| Model     | RMSE  | MSE     | R     | MAE   | MAPE (%) |
|-----------|-------|---------|-------|-------|----------|
| ANN       | 0.046 | 0.00211 | 0.944 | 0.034 | 9.31     |
| MOWAA/ANN | 0.013 | 0.00016 | 0.995 | 0.010 | 2.75     |
| WAA/ANN   | 0.017 | 0.00029 | 0.992 | 0.011 | 3.21     |

Single-objective optimization requires manual weight assignment (e.g.,  $w1$ ) for combining  $RMSE$  and  $R$ , which is arbitrary and dataset-dependent. Multi-objective optimization treats  $RMSE$  and  $R$  as independent goals, automatically exploring their trade-offs without weight bias.

## 2. Complex Trade-offs Captures

Weighted sums may hide conflicts between  $RMSE$  and  $R$  (e.g., improving one degrades the other). Multi-objective methods reveal these trade-offs explicitly via Pareto fronts, allowing flexible post-hoc solution selection.

## 3. Enhances Robustness & Diversity

Single-objective approaches risk local optima (e.g., overfitting  $RMSE$  while ignoring  $R$ ). Multi-objective algorithms preserve solution diversity, balancing:

- High-precision solutions (low  $RMSE$ )
- High-correlation solutions (high  $R$ )
- Compromises (intermediate trade-offs).

To conclude, while multi-objective optimization algorithm introduces additional computational complexity and requires careful parameter tuning, its ability to optimize multiple objectives makes it a powerful tool for ANN hyperparameter optimization, particularly in applications requiring both accuracy and explanatory power.

This study employs the same dataset used in Cheng's research (Cheng and Waele, 2024b) to train, test, and validate the ANN model while maintaining identical network structures and parameters. In previous work, grid search methods were applied to determine the optimal ANN hyperparameters, including the learning rate and number of neurons in the hidden layer, with a single hidden layer proving sufficient for most applications due to the relatively small dataset size (Cheng and Waele, 2024b). The model utilizes *tan - sigmoid* and *pureline* activation functions for the input-hidden and hidden-output layers respectively, with the Levenberg-Marquardt (LM) algorithm employed

for backpropagation training. To ensure robust model performance and mitigate overfitting, K-fold cross-validation ( $K = 5$ ) is implemented, where the original dataset is randomly partitioned into five subsets. Four subsets are used for training while the remaining subset serves for validation. This process is repeated five times such that each subset serves as the validation set exactly once, thereby providing a comprehensive evaluation of model generalization capability.

Based on previous research, the optimized hyperparameters, such as the number of neurons in the hidden layer (set to 6) and the learning rate (set to 0.012), were derived from prior research. Further details are provided in the supplementary material. The MOWAA approach is utilized to iteratively update the weights and thresholds of the ANN model until convergence is achieved. Subsequently, these optimized weights and thresholds are incorporated into the ANN model. For the optimization algorithms, a population size of 100, a maximum of 200 iterations were used, and the size of the Archive set is 100. Therefore, the optimization problem can be defined as expressed in Eq. (20):

$$\begin{aligned} \text{Minimize : } [MSE, (1 - R)^2] &= \text{ANN}(w_{hk}, W_{kl}, b_{hk}, B_{kl}) \\ \text{Variant : } w_{hk}, W_{kl}, b_{hk}, B_{kl} &\in [-0.5, 0.5] \end{aligned} \quad (20)$$

Where  $w_{hk}$  and  $b_{hk}$  are the initial weight and threshold values for the connections between  $h_{th}$  input neurons and  $k_{th}$  hidden neurons, respectively. Besides,  $W_{kl}$  and  $B_{kl}$  are the initial weight and threshold values for the connections between  $k_{th}$  hidden neurons and  $l_{th}$  output neurons, respectively.

The performance of the hybrid MOWAA-ANN in modeling the non-linear relationship between the process parameters and surface waviness of WAAM steel components is illustrated in Fig. 8. From the left graph, the Pareto front indicates that the MOWAA method effectively identifies competitive solutions and stores them in the Archive Set. This demonstrates the model's ability to achieve non-dominated optimization between the two objectives (e.g.,  $MSE$  and  $(1 - R)^2$ ). While it can be observed that some solutions perform better, there is still a conflict between the two objectives in certain cases. Optimizing one objective may lead to a compromise in the other. After the consideration for the values  $((1 - R)^2 + MSE)$ , these values for various non-dominated solutions can be seen in the right graph. Based on the criterion (the minimum  $(1 - R)^2 + MSE$ ), the red spot and the corresponding surrogate model are selected as the potential prediction model in the next section.

The performance of the ANN, WAA-ANN, and MOWAA-ANN models in predicting the non-linear relationship between WAAM process parameters and surface waviness is next compared. Fig. 9 demonstrates that the MOWAA-ANN hybrid model achieves the highest predictive accuracy, with an  $MSE$  of 0.013, followed by WAA-ANN ( $MSE = 0.017$ ) and the unoptimized ANN ( $MSE = 0.046$ ). Furthermore, the prediction errors of the MOWAA-ANN model are consistently within  $\pm 0.03$  mm for all samples, while the WAA-ANN and ANN models exhibit errors outside this range. Fig. 10 highlights the strong correlation between experimental and predicted values of surface waviness, with the MOWAA-ANN model achieving the highest correlation coefficient ( $R = 0.995$ ), followed by WAA-ANN ( $R = 0.992$ ) and ANN ( $R = 0.944$ ).

Finally, Table 20 provides a comprehensive summary of the models' performance using four metrics: root means squared error ( $RMSE$ ), correlation coefficient ( $R$ ), means absolute error ( $MAE$ ), and means absolute percentage error ( $MAPE$ ). Based on the maximum  $R$  and minimum  $MSE$  values, the results clearly demonstrate that the ANN's predictive accuracy improves significantly when optimized using MOWAA or WAA. Moreover, MOWAA outperforms WAA by delivering models with superior generalization, greater flexibility, and a more balanced trade-off between accuracy and explanatory power. These advantages make MOWAA a more effective approach for simultaneously optimizing  $MSE$  and  $R$ , ensuring precise control and optimization of WAAM processes while enhancing the geometrical accuracy and overall quality of components.

## 5. Conclusions

This paper proposes the Multi-Objective Weighted Average Algorithm (MOWAA), a novel extension of the single-objective WAA algorithm that redefines its core mechanics for multi-objective optimization through three key innovations: (1) Hybridized exploitation integrating adaptive mutation and crossover operations to balance exploration-convergence trade-offs; (2) Dynamic archive management with non-dominated sorting across population and elite sets to preserve Pareto diversity; and (3) Fitness-proportional leader selection using roulette wheel to prioritize under-represented regions of the objective space.

To validate the efficacy of MOWAA, it was tested on three series of benchmark test functions (DTLZ, ZDT and CEC2009) and compared with three state-of-the-art optimization algorithms (MOGOA, MOALO, and MOGWO). The proposed algorithm MOWAA displayed an overall better performance than the other algorithms, demonstrating its superiority. Besides, four classical engineering multi-objective optimization problems were applied to evaluate the performance of MOWAA in solving constrained multi-objective optimization problems. Its results also reveal that the MOWAA performs better than the other algorithms. Finally, the MOWAA algorithm was integrated with an ANN model to develop a robust surrogate model for predicting surface waviness in components produced via the wire arc additive manufacturing process. Comprehensive analysis demonstrated that the proposed MOWAA-ANN model significantly outperformed both WAA-ANN and standalone ANN models in terms of predictive accuracy, as reflected by superior metrics such as lower  $RMSE$  and higher  $R$  values. These findings underscore the potential of multi-objective optimization techniques to improve the accuracy and robustness of surrogate models, paving the way for more precise and efficient control of WAAM processes in practical applications. The results obtained in this work collectively highlight the robustness and superior performance of the proposed MOWAA across a broad spectrum of optimization challenges. Therefore, it can be regarded as a reliable alternative for addressing multi-objective optimization problems.

For future works, the application of the proposed algorithm (MOWAA) could be extended to various real-world problems such as multi-objective image segmentation, engineering design problems, etc. Besides, other integration techniques such as the Nelder-Mead method and opposition-based learning can also be considered in an attempt to further enhance the optimization algorithm.

## CRedit authorship contribution statement

**Jun Cheng:** Writing – original draft, Software, Methodology, Investigation, Formal analysis. **Wim De Waele:** Writing – review & editing, Supervision, Resources.

## Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Jun Cheng reports a relationship with China Scholarship Council that includes: funding grants.

Wim De Waele declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgments

Jun Cheng gratefully acknowledge the China Scholarship Council for financial support (No. 770202109210058).

## Appendix C. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.engappai.2025.111569>.

## Appendix. A Detailed description of ZDT and DTLZ benchmark test functions

| Function | Mathematical formulation                                                                                                                                                                                                                                                                                                                                         | D  | Range                                                      |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|------------------------------------------------------------|
| ZDT1     | $F_1 = x_1; F_2 = g \left( 1 - \sqrt{\frac{F_1}{g}} \right); g = 1 + \frac{9}{d-1} \sum_{i=2}^d x_i$                                                                                                                                                                                                                                                             | 30 | $x_i \in [0, 1]$                                           |
| ZDT2     | $F_1 = x_1; F_2 = g \left( 1 - \left( \frac{F_1}{g} \right)^2 \right); g = 1 + \frac{9}{d-1} \sum_{i=2}^d x_i$                                                                                                                                                                                                                                                   | 30 | $x_i \in [0, 1]$                                           |
| ZDT3     | $F_1 = x_1; F_2 = g \left( 1 - \sqrt{F_1 g} - F_{1/g} (10\pi F_1) \right); g = 1 + \frac{9}{d-1} \sum_{i=2}^d x_i$                                                                                                                                                                                                                                               | 30 | $x_i \in [0, 1]$                                           |
| ZDT4     | $F_1 = x_1; F_2 = g \left( 1 - \sqrt{\frac{F_1}{g}} \right); g = 1 + 10(d-1) + \sum_{i=2}^d (x_i^2 - 10 \cos(4\pi x_i))$                                                                                                                                                                                                                                         | 30 | $x_1 \in [0, 1]$<br>$x_i \in [-5, 5]$<br>$i = 2, \dots, D$ |
| DTLZ2    | $F_1 = (1+g) \cos \left( x_1 \left( \frac{\pi}{2} \right) \right) \cos \left( x_2 \left( \frac{\pi}{2} \right) \right); F_2 = (1+g) \cos \left( x_1 \left( \frac{\pi}{2} \right) \right) \sin \left( x_2 \left( \frac{\pi}{2} \right) \right);$<br>$F_3 = (1+g) \sin \left( x_1 \left( \frac{\pi}{2} \right) \right); g = \sum_{i=3}^d (x_i - 0.5)^2;$           | 12 | $x_i \in [0, 1]$                                           |
| DTLZ4    | $F_1 = (1+g) \cos \left( x_1^* \left( \frac{\pi}{2} \right) \right) \cos \left( x_2^* \left( \frac{\pi}{2} \right) \right); F_2 = (1+g) \cos \left( x_1^* \left( \frac{\pi}{2} \right) \right) \sin \left( x_2^* \left( \frac{\pi}{2} \right) \right);$<br>$F_3 = (1+g) \sin \left( x_1^* \left( \frac{\pi}{2} \right) \right); g = \sum_{i=3}^d (x_i - 0.5)^2;$ | 12 | $x_i \in [0, 1]$                                           |

## Appendix B1. Detailed description of the CEC2009 benchmark test functions UF1-UF7

| Function | Mathematical formulation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | D  | Range                                                      |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|------------------------------------------------------------|
| UF1      | $F_1 = x_1 + \frac{2}{ J_1 } \sum_{j \in J_1} \left[ \left( x_j - \sin \left( 6\pi x_1 + \frac{j\pi}{n} \right) \right)^2 \right]; F_2 = 1 - \sqrt{x_1} + \frac{2}{ J_2 } \sum_{j \in J_2} \left[ \left( x_j - \sin \left( 6\pi x_1 + \frac{j\pi}{n} \right) \right) \right]; J_1 = \{j j \text{ is odd and } 2 \leq j \leq n\}; J_2 = \{j j \text{ is even and } 2 \leq j \leq n\}$                                                                                                                                                                                                                                                             | 30 | $x_1 \in [0, 1]$<br>$x_i \in [-1, 1]$<br>$i = 1, \dots, D$ |
| UF2      | $F_1 = x_1 + \frac{2}{ J_1 } \sum_{j \in J_1} y_j^2; F_2 = 1 - \sqrt{x_1} + \frac{2}{ J_2 } \sum_{j \in J_2} y_j^2; J_1 = \{j j \text{ is odd and } 2 \leq j \leq n\}; J_2 = \{j j \text{ is even and } 2 \leq j \leq n\}; y_j =$<br>$\begin{cases} x_j - \left[ \left( 0.3x_1^2 \cos \left( 24\pi x_1 + \frac{4j\pi}{n} \right) + 0.6x_1 \right) \cos \left( 6\pi x_1 + \frac{j\pi}{n} \right) \right] & \text{if } j \in J_1 \\ x_j - \left[ \left( 0.3x_1^2 \cos \left( 24\pi x_1 + \frac{4j\pi}{n} \right) + 0.6x_1 \right) \cos \left( 6\pi x_1 + \frac{j\pi}{n} \right) \right] & \text{if } j \in J_2 \end{cases}$                        | 30 | $x_1 \in [0, 1]$<br>$x_i \in [-1, 1]$<br>$i = 1, \dots, D$ |
| UF3      | $F_1 = x_1 + \frac{2}{ J_1 } \left( 4 \sum_{j \in J_1} y_j^2 - 2 \prod_{j \in J_1} \cos \left( \frac{20y_j\pi}{\sqrt{j}} \right) + 2 \right); F_2 = \sqrt{x_1} + \frac{2}{ J_2 } \left( 4 \sum_{j \in J_2} y_j^2 - 2 \prod_{j \in J_2} \cos \left( \frac{20y_j\pi}{\sqrt{j}} \right) + 2 \right); J_1 \text{ and } J_2 \text{ are the same as those of}$<br>$0.5 \left( 1.0 + \frac{3(j-2)}{\pi-2} \right), j = 2, 3, \dots, n$                                                                                                                                                                                                                  | 30 | $x_i \in [0, 1]$                                           |
| UF4      | $F_1 = x_1 + \frac{2}{ J_1 } \sum_{j \in J_1} h(y_j); F_2 = 1 - x_2 + \frac{2}{ J_2 } \sum_{j \in J_2} h(y_j); J_1 \text{ and } J_2 \text{ are the same as those of UF1, } y_j = x_j - \sin \left( 6\pi x_1 + \frac{j\pi}{n} \right), j = 2, 3, \dots, n; h(t) =$<br>$\frac{ t }{1 + e^{2 t }}$                                                                                                                                                                                                                                                                                                                                                  | 30 | $x_1 \in [0, 1]$<br>$x_i \in [-2, 2]$<br>$i = 2, \dots, D$ |
| UF5      | $F_1 = x_1 + \left( \frac{1}{2N} + \epsilon \right)  \sin(2N\pi x_1)  + \frac{2}{ J_1 } \sum_{j \in J_1} h(y_j); F_2 = 1 - x_1 + \left( \frac{1}{2N} + \epsilon \right)  \sin(2N\pi x_1)  +$<br>$\frac{2}{ J_2 } \sum_{j \in J_2} h(y_j); J_1 \text{ and } J_2 \text{ are identical to those of those of UF1, } y_j = x_j - \sin \left( 6\pi x_1 + \frac{j\pi}{n} \right), j = 2, 3, \dots, n; h(t) = 2t^2 - \cos(4\pi t) + 1$                                                                                                                                                                                                                   | 30 | $x_1 \in [0, 1]$<br>$x_i \in [-1, 1]$<br>$i = 1, \dots, D$ |
| UF6      | $F_1 = x_1 + \max \left\{ 0, 2 \left( \frac{1}{2N} + \epsilon \right)  \sin(2N\pi x_1)  \right\} + \frac{2}{ J_1 } \left( (4 \sum_{j \in J_1} y_j^2 - 2 \prod_{j \in J_1} \cos \left( \frac{20y_j\pi}{\sqrt{j}} \right) + 1) \right); F_2 = 1 - x_1 + \max \left\{ 0, 2 \left( \frac{1}{2N} + \epsilon \right)  \sin(2N\pi x_1)  \right\} + \frac{2}{ J_1 }$<br>$((4 \sum_{j \in J_2} y_j^2 - 2 \prod_{j \in J_2} \cos \left( \frac{20y_j\pi}{\sqrt{j}} \right) + 1) \right); J_1 \text{ and } J_2 \text{ are identical to those of those of UF1, } \epsilon > 0, y_j = x_j - \sin \left( 6\pi x_1 + \frac{j\pi}{n} \right), j = 2, 3, \dots, n$ | 30 | $x_1 \in [0, 1]$<br>$x_i \in [-1, 1]$<br>$i = 1, \dots, D$ |
| UF7      | $F_1 = \sqrt[n]{x_1} + \frac{2}{ J_1 } \sum_{j \in J_1} y_j^2; F_2 = 1 - \sqrt[n]{x_1} + \frac{2}{ J_2 } \sum_{j \in J_2} y_j^2; J_1 \text{ and } J_2 \text{ are identical to those of those of UF1, } \epsilon > 0, y_j = x_j - \sin \left( 6\pi x_1 + \frac{j\pi}{n} \right), j = 2, 3, \dots,$<br>$n$                                                                                                                                                                                                                                                                                                                                         | 30 | $x_1 \in [0, 1]$<br>$x_i \in [-1, 1]$<br>$i = 1, \dots, D$ |

**Appendix B2. Detailed description of the CEC2009 benchmark test functions UF8-UF10**

| Function | Mathematical formulation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | D  | Range                                                                          |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|--------------------------------------------------------------------------------|
| UF8      | $F_1 = \cos(0.5x_1\pi)\cos(0.5x_2\pi) + \frac{2}{ J_1 } \sum_{j \in J_1} \left( x_j - 2x_2 \sin\left(2\pi x_1 + \frac{j\pi}{n}\right)^2 \right);$ $F_2 = \cos(0.5x_1\pi)\sin(0.5x_2\pi) + \frac{2}{ J_2 } \sum_{j \in J_2} \left( x_j - 2x_2 \sin\left(2\pi x_1 + \frac{j\pi}{n}\right)^2 \right);$ $F_3 = \sin(0.5x_1\pi) + \frac{2}{ J_3 } \sum_{j \in J_3} \left( x_j - 2x_2 \sin\left(2\pi x_1 + \frac{j\pi}{n}\right)^2 \right);$ $J_1 = \{j 3 \leq j \leq n, \text{ and } j-1 \text{ is a multiplication of } 3\};$ $J_2 = \{j 3 \leq j \leq n, \text{ and } j-2 \text{ is a multiplication of } 3\};$ $J_3 = \{j 3 \leq j \leq n, \text{ and } j \text{ is a multiplication of } 3\};$                                                                    | 30 | $x_1 \in [0, 1]$<br>$x_2 \in [0, 1]$<br>$x_i \in [-2, 2]$<br>$i = 1, \dots, D$ |
| UF9      | $F_1 = 0.5[\max\{0, (1 + \varepsilon)(1 - 4(2x_1 - 1)^2)\}] + \frac{2}{ J_1 } \sum_{j \in J_1} \left( x_j - 2x_2 \sin\left(2\pi x_1 + \frac{j\pi}{n}\right)^2 \right);$ $F_2 = 0.5[\max\{0, (1 + \varepsilon)(1 - 4(2x_1 - 1)^2)\} + 2x_1 x_2 ] + \frac{2}{ J_2 } \sum_{j \in J_2} \left( x_j - 2x_2 \sin\left(2\pi x_1 + \frac{j\pi}{n}\right)^2 \right);$ $F_3 = 1 - x_2 + \frac{2}{ J_3 } \sum_{j \in J_3} \left( x_j - 2x_2 \sin\left(2\pi x_1 + \frac{j\pi}{n}\right)^2 \right);$ $J_1 = \{j 3 \leq j \leq n, \text{ and } j-1 \text{ is a multiplication of } 3\};$ $J_2 = \{j 3 \leq j \leq n, \text{ and } j-2 \text{ is a multiplication of } 3\};$ $J_3 = \{j 3 \leq j \leq n, \text{ and } j \text{ is a multiplication of } 3\}, \varepsilon = 0.1;$ | 30 | $x_1 \in [0, 1]$<br>$x_2 \in [0, 1]$<br>$x_i \in [-2, 2]$<br>$i = 1, \dots, D$ |
| UF10     | $F_1 = \cos(0.5x_1\pi)\cos(0.5x_2\pi) + \frac{2}{ J_1 } \sum_{j \in J_1} (4y_j^2 - \cos(8\pi y_j) + 1);$ $F_2 = \cos(0.5x_1\pi)\sin(0.5x_2\pi) + \frac{2}{ J_2 } \sum_{j \in J_2} (4y_j^2 - \cos(8\pi y_j) + 1);$ $F_3 = \sin(0.5x_1\pi) + \frac{2}{ J_3 } \sum_{j \in J_3} (4y_j^2 - \cos(8\pi y_j) + 1);$ $J_1 = \{j 3 \leq j \leq n, \text{ and } j-1 \text{ is a multiplication of } 3\};$ $J_2 = \{j 3 \leq j \leq n, \text{ and } j-2 \text{ is a multiplication of } 3\};$ $J_3 = \{j 3 \leq j \leq n, \text{ and } j \text{ is a multiplication of } 3\};$                                                                                                                                                                                               | 30 | $x_1 \in [0, 1]$<br>$x_2 \in [0, 1]$<br>$x_i \in [-2, 2]$<br>$i = 1, \dots, D$ |

**Appendix. C Detailed description of the engineering problems****1. BNH problem:**

$$\text{Minimize: } f_1(x) = 4x_1^2 + 4x_2^2$$

$$\text{Minimize: } f_2(x) = (x_1 - 5)^2 + (x_2 - 5)^2$$

$$\text{Constraint: } g_1(x) = (x_1 - 5)^2 + x_2^2 - 25; g_2(x) = 7.7 - (x_1 - 8)^2 + (x_2 + 3)^2$$

$$\text{Variant range: } 0 \leq x_1 \leq 5, 0 \leq x_3 \leq 3$$

**2. CONSTR:**

$$\text{Minimize: } f_1(x) = x_1$$

$$\text{Minimize: } f_2(x) = (1 + x_2)/x_1$$

$$\text{Constraint: } g_1(x) = 6 - (x_2 + 9x_1); g_2(x) = 1 + (x_2 - 9x_1)$$

$$\text{Variant range: } 0.1 \leq x_1 \leq 1, 0 \leq x_3 \leq 5$$

**3. SRN:**

$$\text{Minimize: } f_1(x) = 2 + (x_1 - 2)^2 + (x_2 - 1)^2$$

$$\text{Minimize: } f_2(x) = 9x_1 - (x_2 - 1)^2$$

$$\text{Constraint: } g_1(x) = x_1^2 + x_2^2 - 255; g_2(x) = x_1 - 3x_2 + 10$$

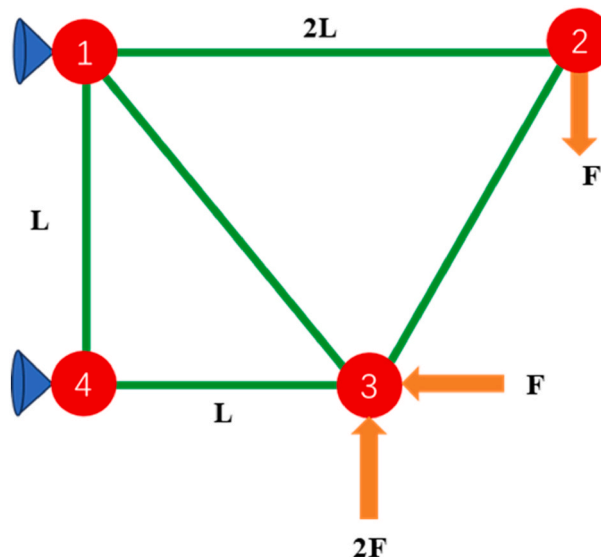
$$\text{Variant range: } -20 \leq x_1 \leq 20, -20 \leq x_3 \leq 20$$

**4. BAR TRUSS:**

$$\text{Minimize: } f_1(x) = 200 \times (2 \times x_1 + \sqrt{2} \times x_2) + \sqrt{2} \times x_3 + x_4$$

$$\text{Minimize: } f_2(x) = 0.01 \times \frac{2}{x_1} + \frac{2 \times \sqrt{2}}{x_2} - \frac{2 \times \sqrt{2}}{x_3} + \frac{2}{x_4}$$

$$\text{Variant range: } 1 \leq x_1 \leq 3; 1.4142 \leq x_2 \leq 3; 1.4142 \leq x_3 \leq 3; 1 \leq x_4 \leq 3;$$



## Data availability

Data will be made available on request.

## References

- Armano, G., Farmani, M.R., 2016. Multiobjective clustering analysis using particle swarm optimization. *Expert Syst. Appl.* 55, 184–193. <https://doi.org/10.1016/j.eswa.2016.02.009>.
- Chen, S., Pan, Q., Gao, L., Sang, H., 2021. A population-based iterated greedy algorithm to minimize total flowtime for the distributed blocking flowshop scheduling problem. *Eng. Appl. Artif. Intell.* 104, 104375–104388. <https://doi.org/10.1016/j.engappai.2021.104375>.
- Cheng, J., Waele, W. De, 2024a. Weighted average algorithm : a novel meta-heuristic optimization algorithm based on the weighted average position concept. *Knowl. Base Syst.* 305, 112564–112593. <https://doi.org/10.1016/j.knsys.2024.112564>.
- Cheng, J., Waele, W. De, 2024b. Prediction and optimization of surface waviness of WAAM components using a hybrid Rank-Gaussian PSO algorithm and ANN. *Structures* 69, 107247–107265. <https://doi.org/10.1016/j.istruc.2024.107247>.
- D'Angelo, G., Palmieri, F., 2021. GGA: a modified genetic algorithm with gradient-based local search for solving constrained optimization problems. *Inf. Sci.* 547, 136–162. <https://doi.org/10.1016/j.ins.2020.08.040>.
- De Farias, L.R.C., Bassani, H.F., Braga, P.H.M., Araújo, A.F.R., 2018. MOEA/D with uniformly randomly adaptive weights. In: *GECCO 2018 - Proc. 2018 Genet. Evol. Comput. Conf.*, pp. 641–648. <https://doi.org/10.1145/3205455.3205648>.
- Deb, K., 2012. Advances in evolutionary multi-objective optimization. *Lect. Notes Comput. Sci.* 7515, 1–26. [https://doi.org/10.1007/978-3-642-33119-0\\_1](https://doi.org/10.1007/978-3-642-33119-0_1).
- Deb, K., Jain, H., 2014. An evolutionary many-objective optimization algorithm using reference-point-based nondominated sorting approach, Part I: solving problems with box constraints. *IEEE Trans. Evol. Comput.* 18, 577–601. <https://doi.org/10.1109/TEVC.2013.2281535>.
- Ding, H., Sun, Y., Huang, N., Shen, Z., Wang, Z., Iftekhar, A., Cui, X., 2023. RVGAN-TL: a generative adversarial networks and transfer learning-based hybrid approach for imbalanced data classification. *Inf. Sci.* 629, 184–203. <https://doi.org/10.1016/j.ins.2023.01.147>.
- González, M.A., Rasconi, R., Oddi, A., 2022. Metaheuristics for multiobjective optimization in energy-efficient job shops. *Eng. Appl. Artif. Intell.* 115, 105263–105284. <https://doi.org/10.1016/j.engappai.2022.105263>.
- Gunantara, N., 2018. A review of multi-objective optimization: methods and its applications. *Cogent Eng.* 5, 1–16. <https://doi.org/10.1080/23311916.2018.1502242>.
- Hong, M.-H., Witt, J.K., Szafr, D.A., 2022. The weighted average illusion: biases in perceived mean position in scatterplots. *IEEE Trans. Vis. Comput. Graph.* 28, 987–997. <https://doi.org/10.1109/TVCG.2021.3114783>.
- Hu, X., Guo, P., Li, Y., Li, G., Cui, Z., Yang, J., 2024. TVDO: Tchebycheff value-decomposition optimization for multiagent reinforcement learning. *IEEE Transact. Neural Networks Learn. Syst.* 1–14. <https://doi.org/10.1109/TNNLS.2024.3455422>.
- Khodadadi, N., Abualigah, L., Al-Tashi, Q., Mirjalili, S., 2023. Multi-objective chaos game optimization. *Neural Comput. Appl.* 35, 14973–15004. <https://doi.org/10.1007/s00521-023-08432-0>.
- Kumar, S., Jangir, P., Tejani, G.G., Premkumar, M., 2022. A decomposition based multi-objective heat transfer search algorithm for structure optimization. *Knowl. Base Syst.* 253, 109591–109620. <https://doi.org/10.1016/j.knsys.2022.109591>.
- Kumar, S., Tejani, G.G., Pholdee, N., Bureerat, S., 2021a. Multiobjective structural optimization using improved heat transfer search. *Knowl. Base Syst.* 219, 106811–106831. <https://doi.org/10.1016/j.knsys.2021.106811>.
- Kumar, S., Tejani, G.G., Pholdee, N., Bureerat, S., Mehta, P., 2021b. Hybrid heat transfer search and passing vehicle search optimizer for multi-objective structural optimization. *Knowl. Base Syst.* 212, 106556–106574. <https://doi.org/10.1016/j.knsys.2020.106556>.
- Li, G., Wang, G.G., Xiao, R. Bin, 2022. A novel adaptive weight algorithm based on decomposition and two-part update strategy for many-objective optimization. *Inf. Sci.* 615, 323–347. <https://doi.org/10.1016/j.ins.2022.09.057>.
- Li, H., Landa-Silva, D., 2008. Evolutionary multi-objective simulated annealing with adaptive and competitive search direction. In: *2008 IEEE Congr. Evol. Comput. CEC 2008*, pp. 3311–3318. <https://doi.org/10.1109/CEC.2008.4631246>.
- Li, W., Zeng, X., Huang, Y., Cheung, Y., 2025. HK-MOEA/D : a historical knowledge-guided resource allocation for decomposition multiobjective optimization. *Eng. Appl. Artif. Intell.* 139, 109482–109497. <https://doi.org/10.1016/j.engappai.2024.109482>.
- Luo, T., Xie, J., Zhang, B., Zhang, Y., Li, C., Zhou, J., 2024. An improved levy chaotic particle swarm optimization algorithm for energy-efficient cluster routing scheme in industrial wireless sensor networks. *Expert Syst. Appl.* 241, 122780–122800. <https://doi.org/10.1016/j.eswa.2023.122780>.
- Ma, X., Yu, Y., Li, X., Qi, Y., Zhu, Z., 2020. A survey of weight vector adjustment methods for decomposition-based multiobjective evolutionary algorithms. *IEEE Trans. Evol. Comput.* 24, 634–649. <https://doi.org/10.1109/TEVC.2020.2978158>.
- Mardle, S., Miettinen, K.M., 2000. Nonlinear multiobjective optimization. *J. Oper. Res. Soc.* 51, 246–330. <https://doi.org/10.1023/254267>.
- Ming, F., Gong, W., Zhen, H., Wang, L., Gao, L., 2024. Constrained multi-objective optimization evolutionary algorithm for real-world continuous mechanical design problems. *Eng. Appl. Artif. Intell.* 135, 108673–108690. <https://doi.org/10.1016/j.engappai.2024.108673>.
- Minh, H., Sang-to, T., Abdel, M., Cuong-le, T., 2022. A new metaheuristic optimization based on K-means clustering algorithm and its application to structural damage identification. *Knowl. Base Syst.* 251, 109189–109224. <https://doi.org/10.1016/j.knsys.2022.109189>.
- Mirjalili, S., Saremi, S., Mirjalili, S.M., Coelho, L.D.S., 2016. Multi-objective grey wolf optimizer: a novel algorithm for multi-criterion optimization. *Expert Syst. Appl.* 47, 106–119. <https://doi.org/10.1016/j.eswa.2015.10.039>.
- Mohamed, H.S., Qiong, T., Isleem, H.F., Tipu, R.K., Shahin, R.I., Yehia, S.A., Jangir, P., Arpita, Khishe, M., 2024. Compressive behavior of elliptical concrete-filled steel tubular short columns using numerical investigation and machine learning techniques. *Sci. Rep.* 14, 27007–27038. <https://doi.org/10.1038/s41598-024-77396-5>.
- Pereira, J.L.J., Oliver, G.A., Francisco, M.B., Cunha, S.S., Gomes, G.F., 2022. A review of multi-objective optimization: methods and algorithms in mechanical engineering problems. *Arch. Comput. Methods Eng.* 29, 2285–2308. <https://doi.org/10.1007/s11831-021-09663-x>.
- Qi, Y., Liu, D., Li, X., Lei, J., Xu, X., Miao, Q., 2020. An adaptive penalty-based boundary intersection method for many-objective optimization problem. *Inf. Sci.* 509, 356–375. <https://doi.org/10.1016/j.ins.2019.03.040>.

- Qin, D., He, Z., Zhao, X., Liu, J., Zhang, F., Xiao, L., 2023. Area and power optimization for fixed polarity reed – muller logic circuits based on multi-strategy multi-objective artificial bee colony algorithm. *Eng. Appl. Artif. Intell.* 121, 105906–105917. <https://doi.org/10.1016/j.engappai.2023.105906>.
- Qin, M., Zhang, C., Bai, B., Zhang, G., Yeung, D.Y., 2023. High-quality temporal link prediction for weighted dynamic graphs via inductive embedding aggregation. *IEEE Trans. Knowl. Data Eng.* 35, 9378–9393. <https://doi.org/10.1109/TKDE.2023.3238360>.
- Schaffer, J.D., 1985. Multiple objective optimization with vector evaluated genetic algorithms. In: *Proc. First Int. Conf. Genet. Algorithms Their Appl.*, pp. 93–100.
- Sharma, S., Kumar, V., 2022. A comprehensive review on multi-objective optimization techniques: past, present and future. *Arch. Comput. Methods Eng.* 29, 5605–5633. <https://doi.org/10.1007/s11831-022-09778-9>.
- Srikanth, G., Nimma, D., Lalitha, R.V.S., Jangir, P., Kumari, N.V.S., Arpita, 2024. Food security based marine life ecosystem for polar region conditioning: remote sensing analysis with machine learning model. *Remote Sens. Earth Syst. Sci.* 8, 65–73. <https://doi.org/10.1007/s41976-024-00172-6>.
- Talatahari, S., Azizi, M., 2020. Optimization of constrained mathematical and engineering design problems using chaos game optimization. *Comput. Ind. Eng.* 145, 106560–106589. <https://doi.org/10.1016/j.cie.2020.106560>.
- Tan, K.C., Yang, Y.J., Goh, C.K., 2006. A distributed cooperative coevolutionary algorithm for multiobjective optimization. *IEEE Trans. Evol. Comput.* 10, 527–549. <https://doi.org/10.1109/TEVC.2005.860762>.
- Wang, P., Xiao, H., Han, X., Yang, F., Li, L., 2023. A coevolutionary algorithm based on reference line guided archive for constrained multiobjective optimization. *Appl. Soft Comput.* 142, 110169–110180. <https://doi.org/10.1016/j.asoc.2023.110169>.
- Wang, X., Zhang, X., Qin, B., Guo, L., 2023. Improved multi-objective grasshopper optimization algorithm and application in capacity configuration of urban rail hybrid energy storage systems. *J. Energy Storage* 72, 108363–108386. <https://doi.org/10.1016/j.est.2023.108363>.
- Wolpert, D.H., Macready, W.G., 1997. No free lunch theorems for optimization. *IEEE Trans. Evol. Comput.* 1, 67–82. <https://doi.org/10.1109/4235.585893>.
- Yacoubi, S., Manita, G., Chhabra, A., Korbaa, O., Mirjalili, S., 2023. A multi-objective Chaos Game Optimization algorithm based on decomposition and random learning mechanisms for numerical optimization. *Appl. Soft Comput.* 144, 110525–110565. <https://doi.org/10.1016/j.asoc.2023.110525>.
- Yen, G.G., Lu, H., 2003. Dynamic multiobjective evolutionary algorithm: adaptive cell-based rank and density estimation. *IEEE Trans. Evol. Comput.* 7, 253–274. <https://doi.org/10.1109/TEVC.2003.810068>.
- Yue, C., Isleem, H.F., Qader, D.N., Mahmoudian, A., Weiming, T., Jangir, P., Arpita, Hamed, A.Y., 2024. Utilization finite element and machine learning methods to investigation the axial compressive behavior of elliptical FRP-confined concrete columns. *Structures* 70, 107681–107714. <https://doi.org/10.1016/j.istruc.2024.107681>.
- Zapotecas-martínez, S., García-nájera, A., Menchaca-méndez, A., 2023. Engineering applications of multi-objective evolutionary algorithms : a test suite of box-constrained real-world problems. *Eng. Appl. Artif. Intell.* 123, 106192–106212. <https://doi.org/10.1016/j.engappai.2023.106192>.
- Zhang, T., Qi, W., Zhao, X., Yan, Y., Cao, Y., 2022. A local dimming method based on improved multi-objective evolutionary algorithm. *Expert Syst. Appl.* 204, 2285–2308. <https://doi.org/10.1016/j.eswa.2022.117468>.